



Generalized Rank-based Evaluation for Knowledge Graph Completion : Challenges, Framework, and Analyses

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Under review

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■ Combining Perspectives

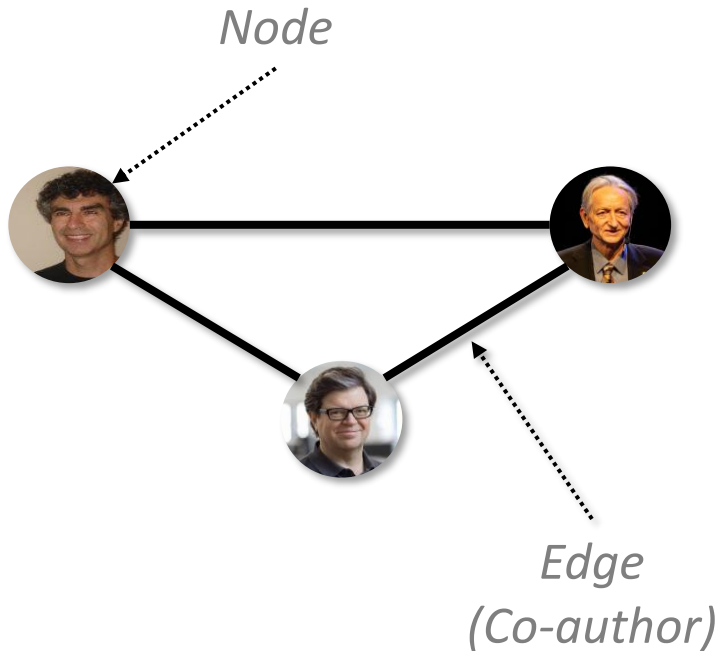
■ Experiments

■ Conclusion & Future Works

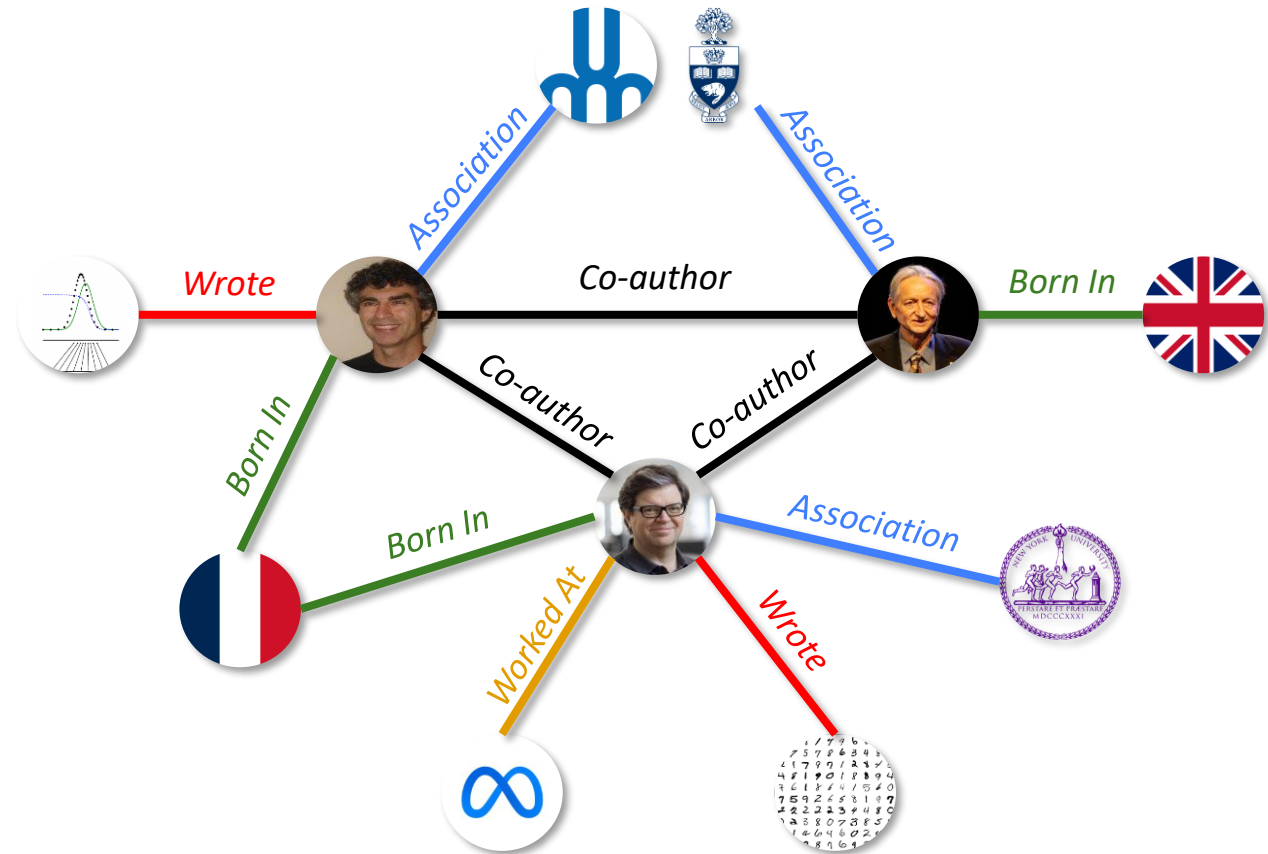
KNOWLEDGE GRAPH COMPLETION

■ What is a Knowledge Graph (KG)?

[Ordinary Graph]



[Knowledge Graph]

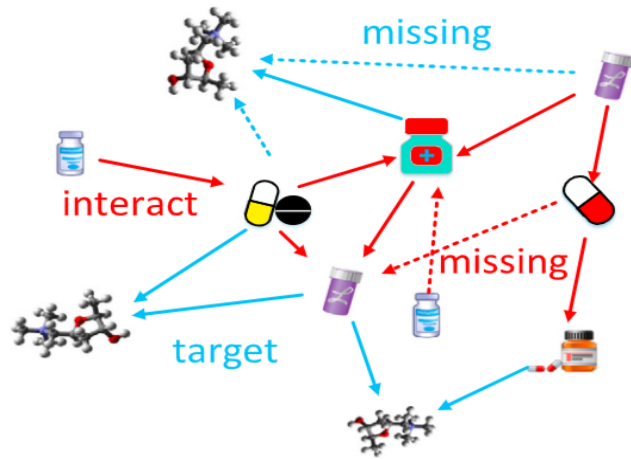


KNOWLEDGE GRAPH COMPLETION

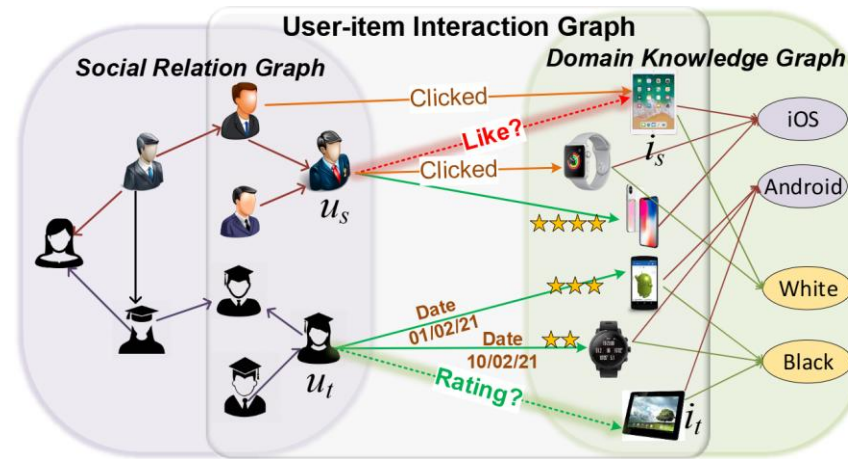
■ Multi-relationship Between Entities

- Edge : **Triple** (head entity, relation, tail entity)
- Rich representation → applied in various domains

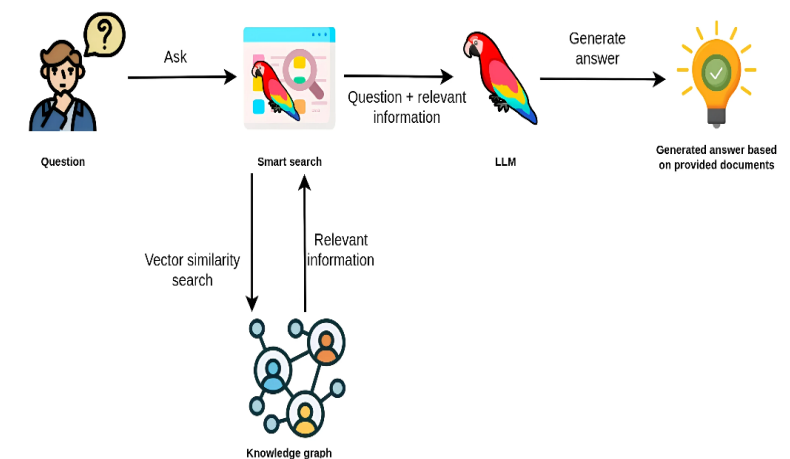
<Drug Discovery>



<Recommender System>



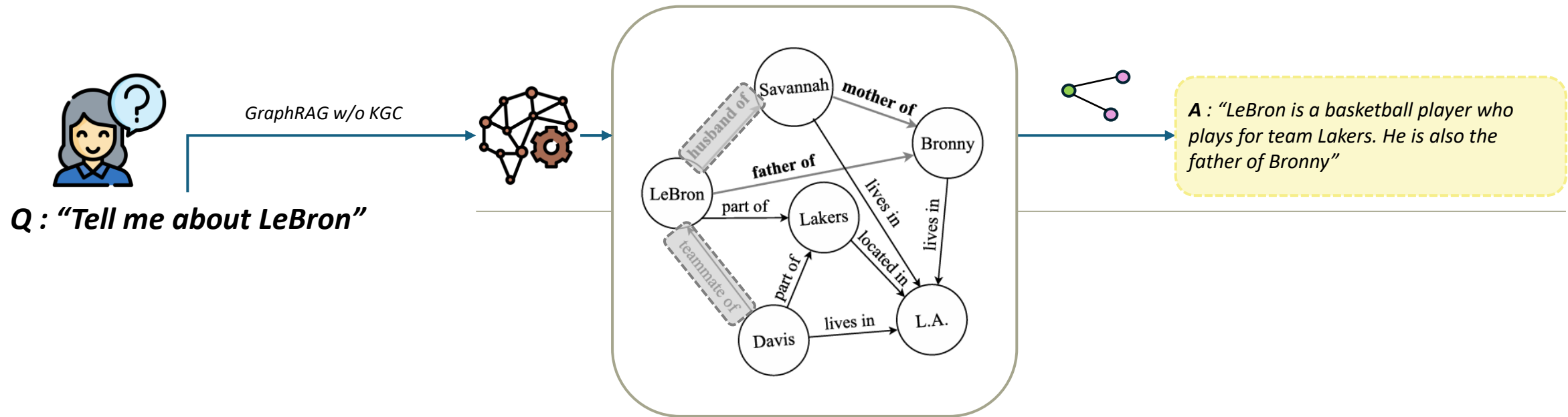
<GraphRAG>



KNOWLEDGE GRAPH COMPLETION

■ Missing Facts in KGs

- In Freebase and DBpedia, more than 66% of the person entities are missing a birthplace
- Greatly hinders the performance of systems that rely upon it

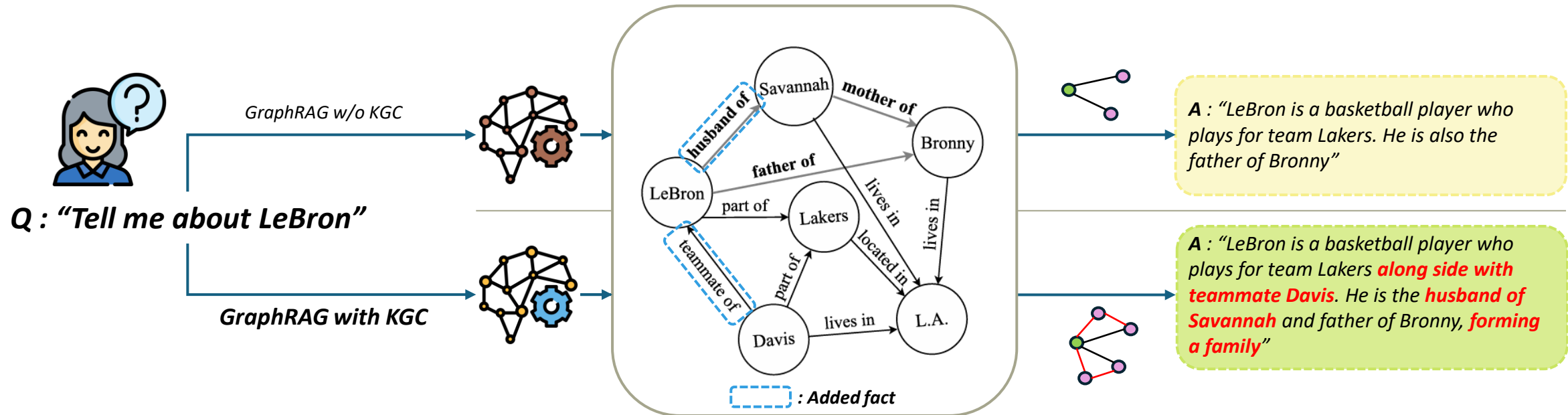


KNOWLEDGE GRAPH COMPLETION

■ Knowledge Graph Completion (KGC)

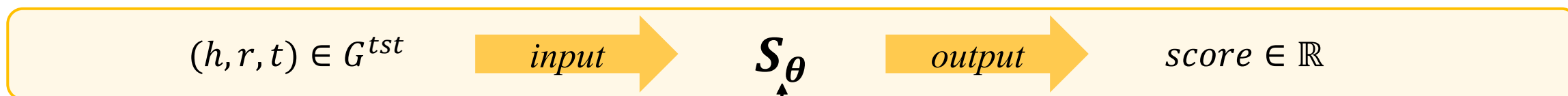
- KGC model to automatically fill missing links has been extensively studied
- Normally two tasks exist, link prediction(entity prediction) and relation prediction

link prediction : $(h, r, ?)$ or $(?, r, t) \rightarrow \text{predict "?"}$

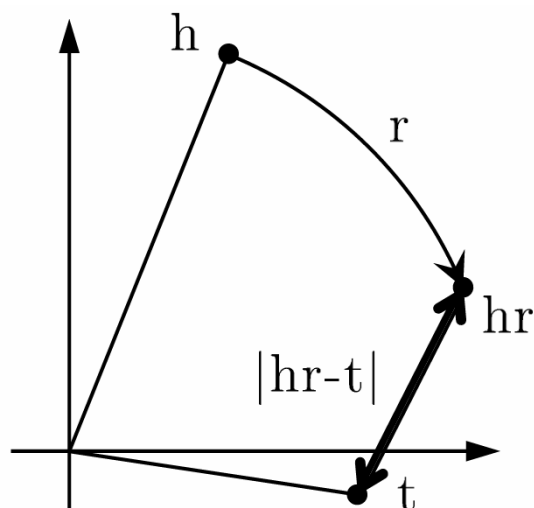


KNOWLEDGE GRAPH COMPLETION

KGC Expression & Categories

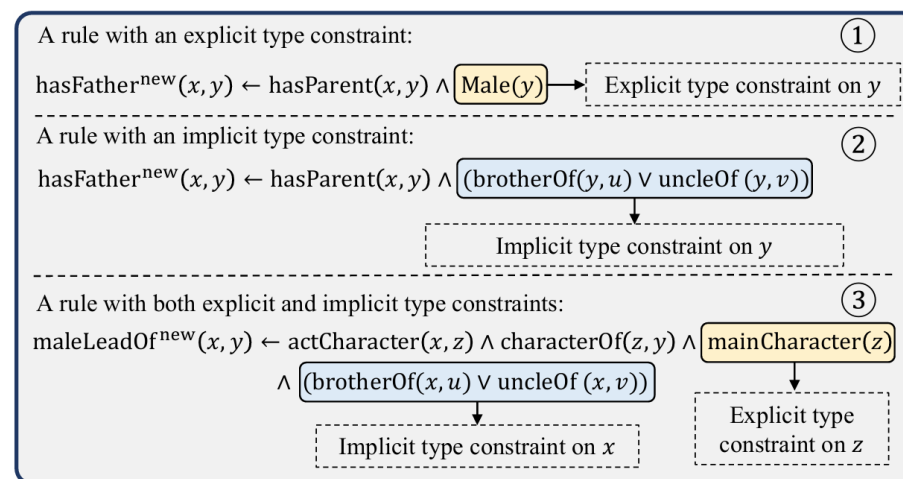


Embedding Based



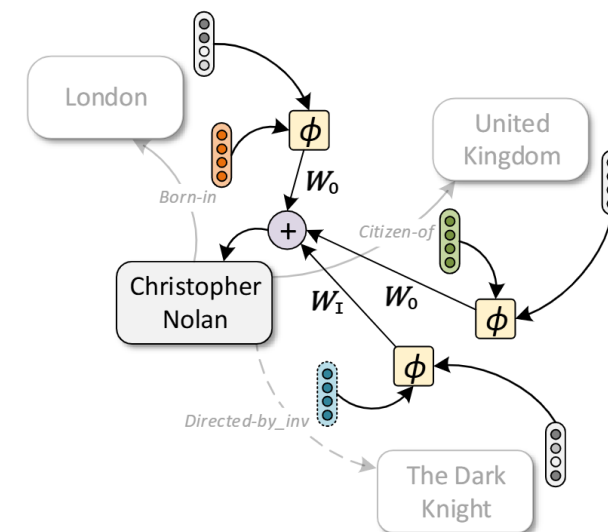
RotatE (ICLR'19)

Rule-Mining Based



TCLM (CIKM'24)

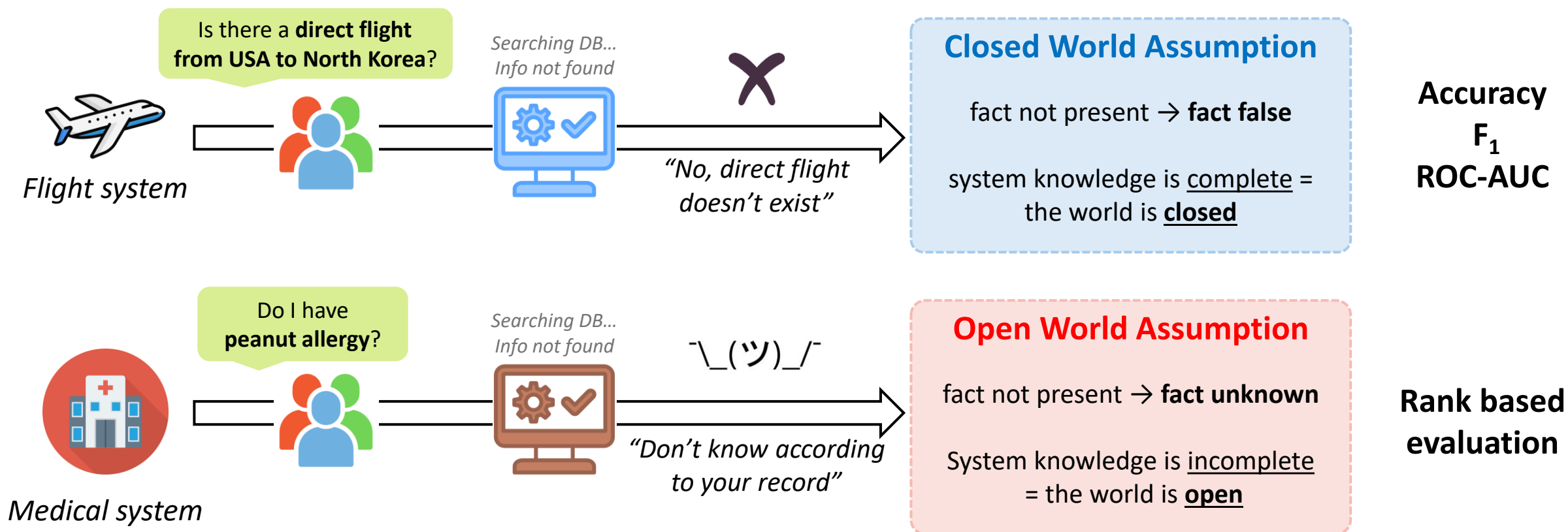
GNN Based



CompGCN (NeurIPS'20)

KGC EVALUATION & MOTIVATION

■ Inherent and Philosophical Question

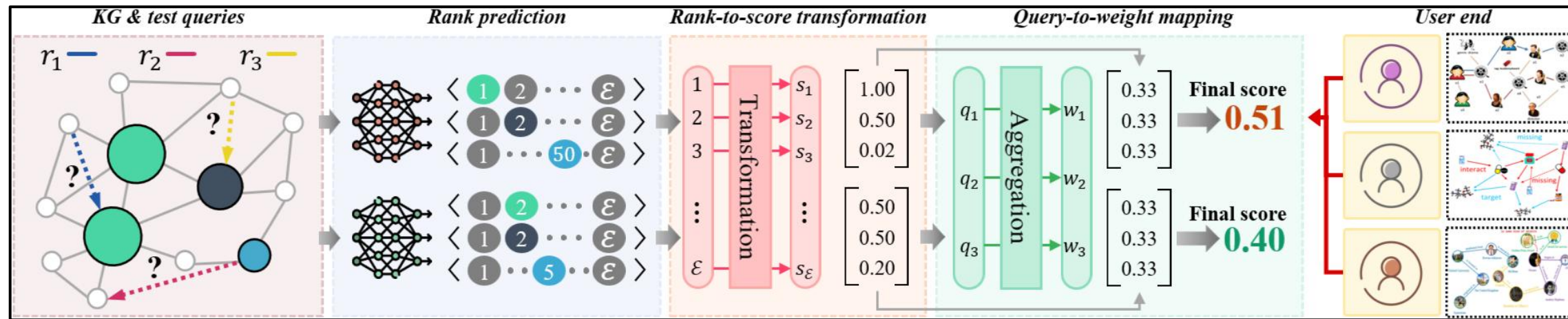


KGC EVALUATION & MOTIVATION

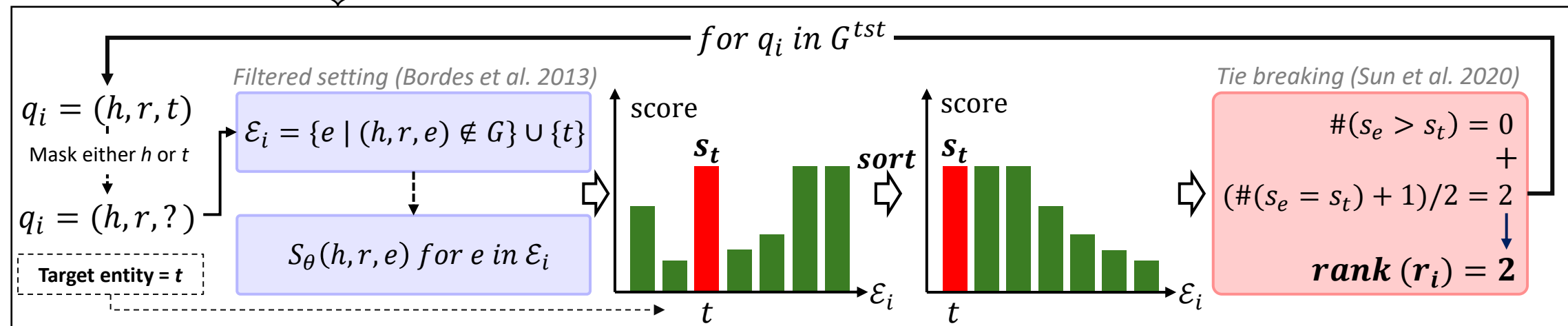
Overview of Rank Based Metrics

$(h, r, t) \in G^{tst}$ $\xrightarrow{\text{input}}$ S_θ $\xrightarrow{\text{output}}$ $\text{score} \in \mathbb{R}$

Metric Calc.



Rank Prediction

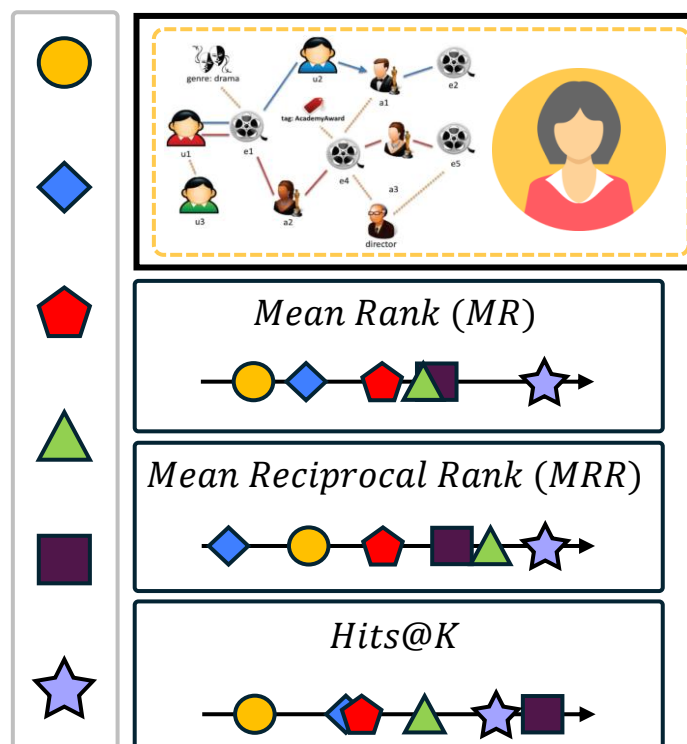


KGC EVALUATION & MOTIVATION

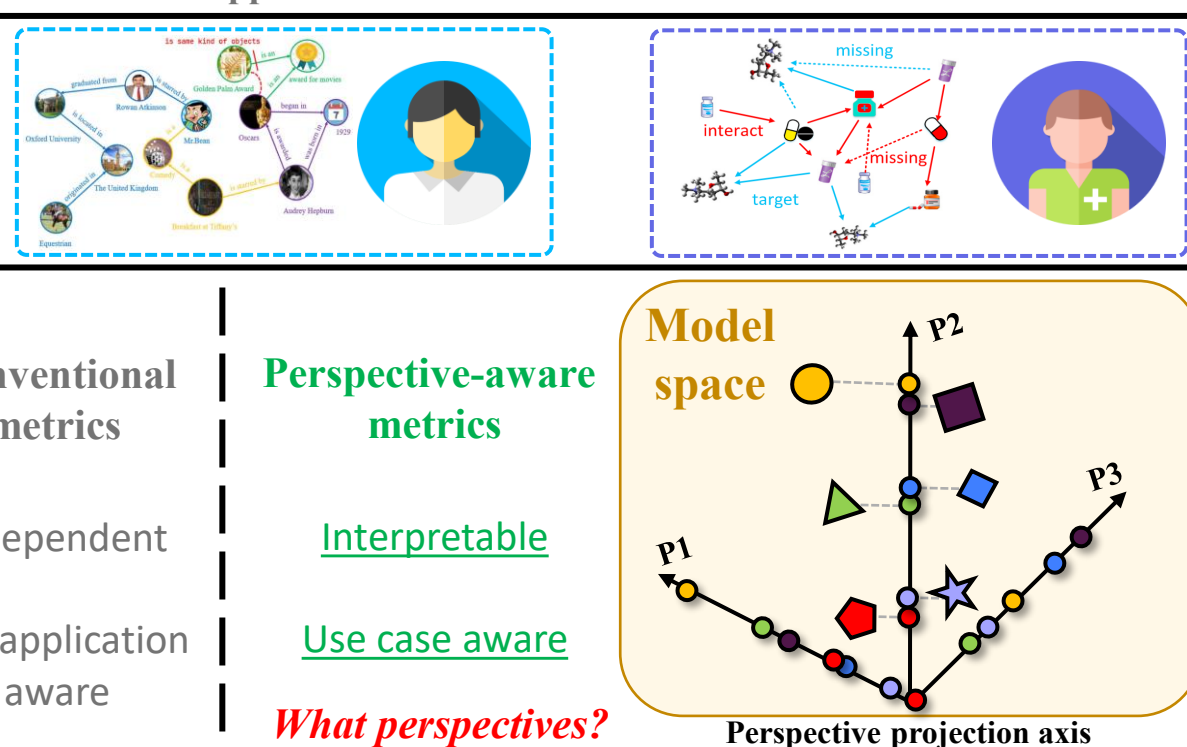
Shortcomings of Conventional Metrics

- Each practitioner has different characteristic, use of KG
- Multi-perspective evaluation is crucial for application-aware model selection

Models



Applications



Conventional
metrics

Independent

Not application
aware

Perspective-aware
metrics

Interpretable

Use case aware

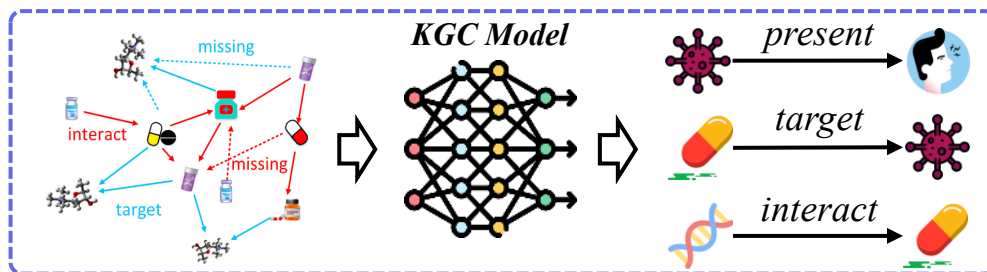
What perspectives?

Model
space

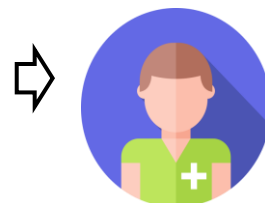
Perspective projection axis

KGC EVALUATION & MOTIVATION

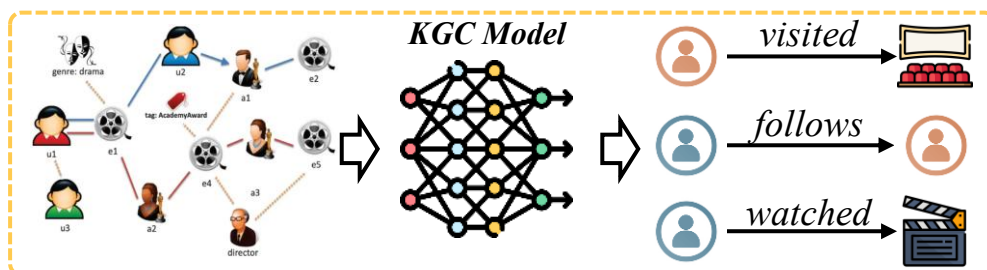
■ (P1) Predictive Sharpness



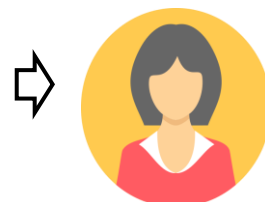
Drug prediction



“Clinical trials take years and millions of USD [1],
we can't tolerate less faithful facts”
*Mostly interested in **top-rank prediction***

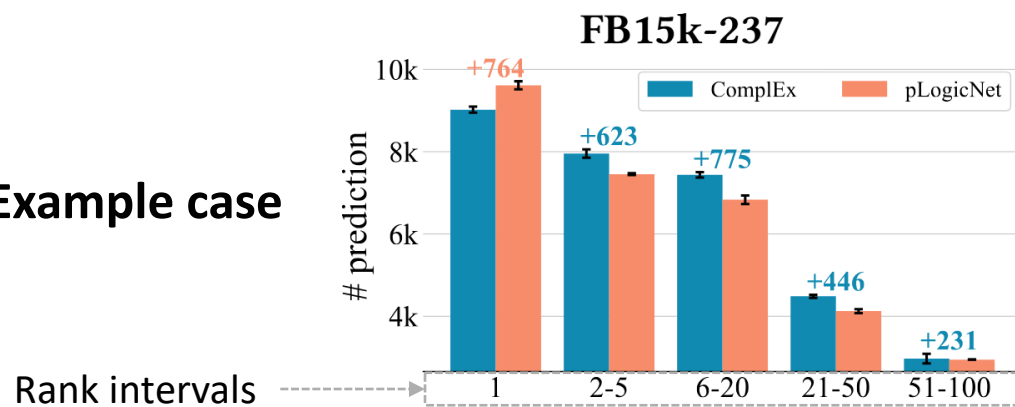


Rec. system



“As long as the KG captures meaningful relationship
between user and item, the fact is acceptable”
*More generous to **less likely prediction***

Example case



pLogicNet

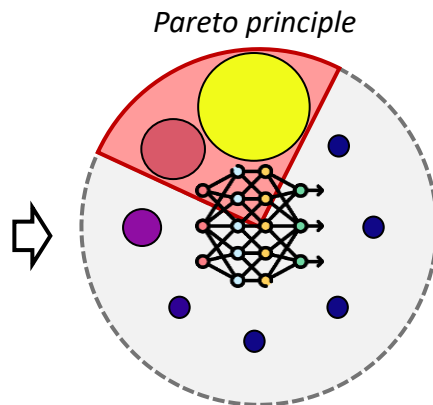
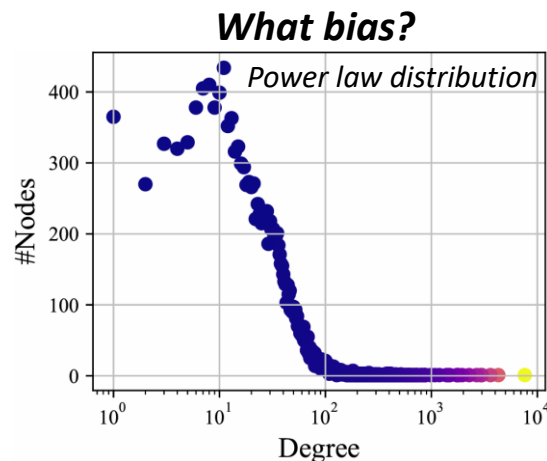


ComplEx



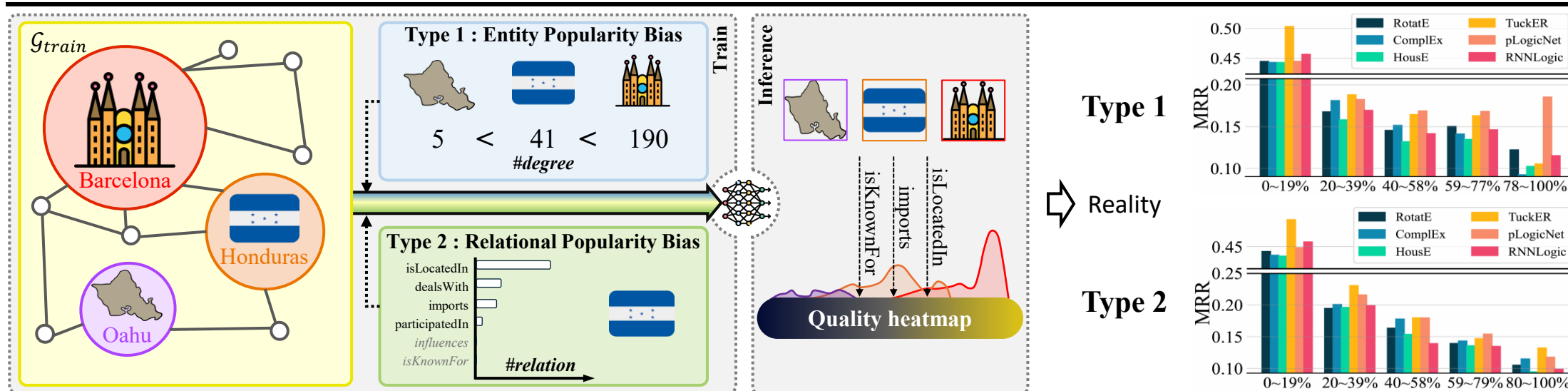
KGC EVALUATION & MOTIVATION

■ (P2) Popularity Bias Robustness



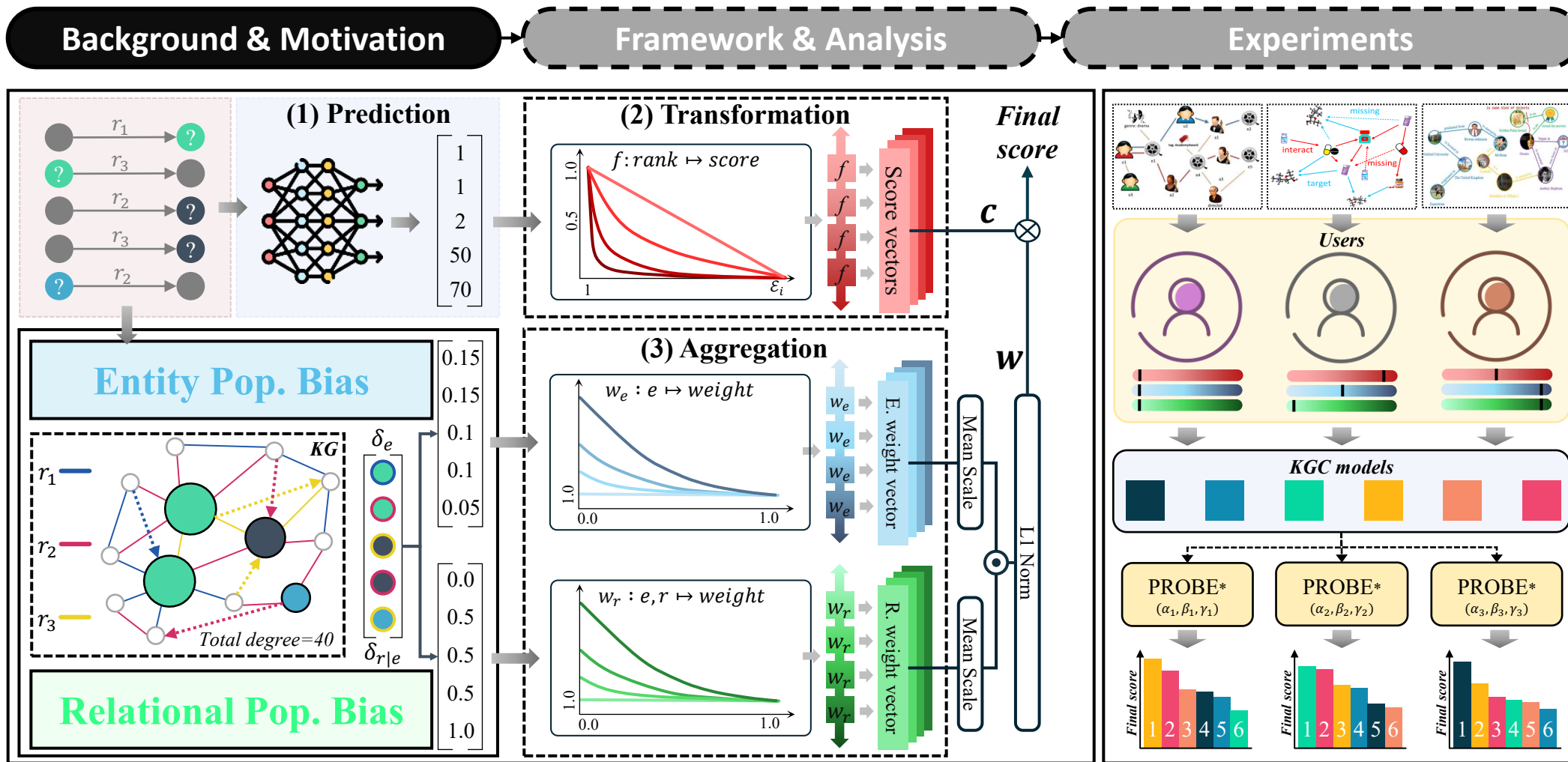
"We treat all facts equally."
Interested in **pure precision**

"Fresh and not so obvious facts convey more value"
Interested in discovering **minor facts**



KGC EVALUATION & MOTIVATION

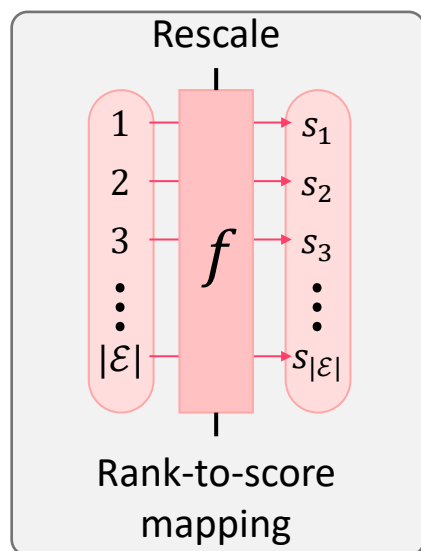
Overview of PROBE*



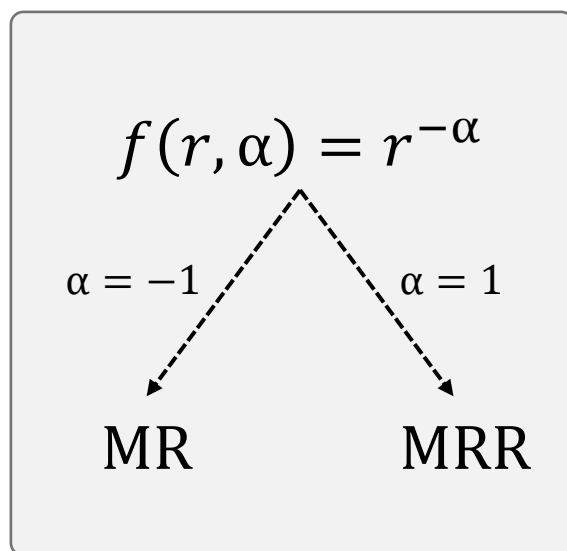
PREDICTIVE SHARPNESS

Rank Transformation (f) : Scoring Ranks

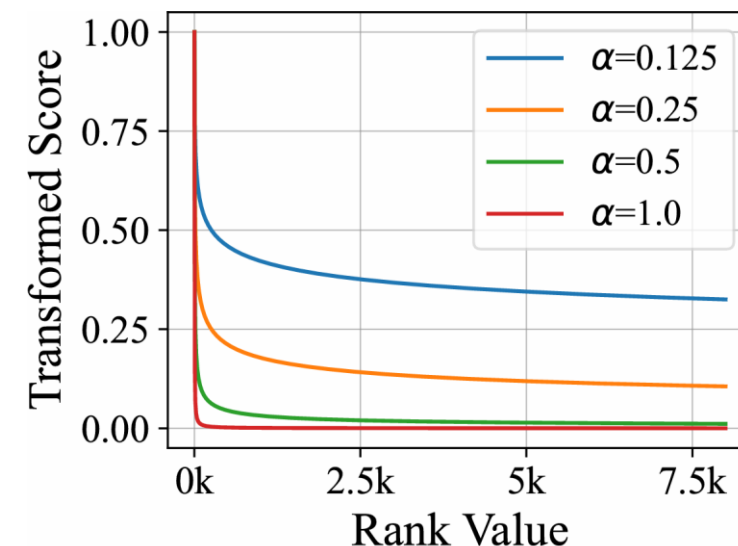
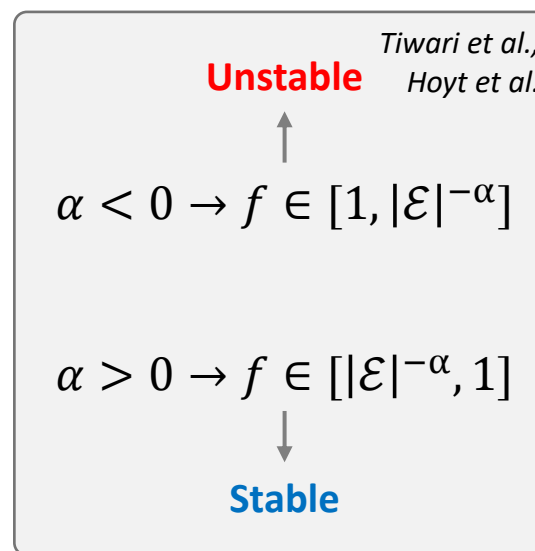
Rank Transformation



Sharpness control factor (α)



Evaluation



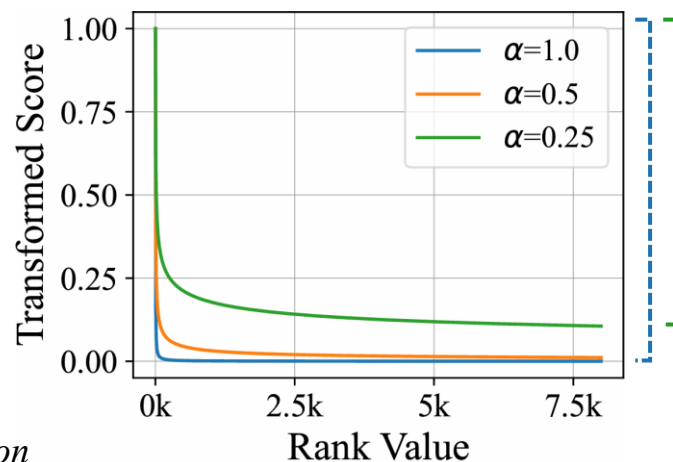
[1] Tiwari, Sudhanshu, Iti Bansal, and Carlos R. Rivero. "Revisiting the evaluation protocol of knowledge graph completion methods for link prediction." *Proceedings of the Web Conference 2021*.

[2] Hoyt, Charles Tapley, et al. "A unified framework for rank-based evaluation metrics for link prediction in knowledge graphs." *arXiv preprint arXiv:2203.07544* (2022).

PREDICTIVE SHARPNESS

■ Subtle, Yet Crucial Details

$$f(r, \alpha) = r^{-\alpha}$$



Problem

Compressed upper and lower bound

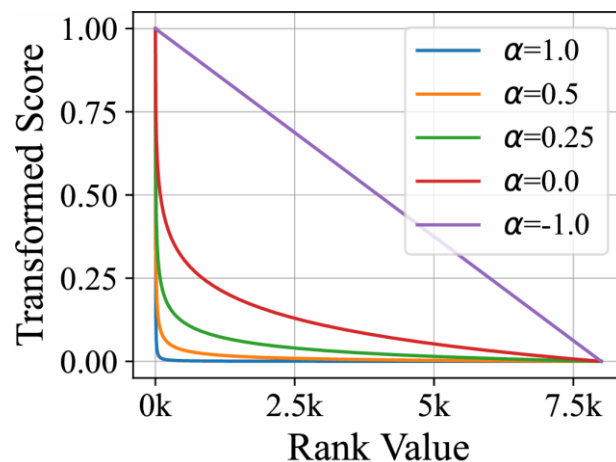


Solution

Securing absolute lower bound with ϵ_i

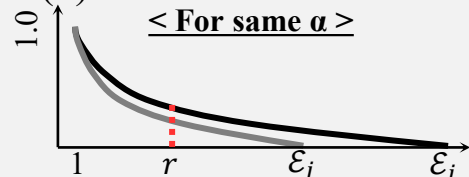
Affine transformation

$$f^*(r, \alpha) = \frac{r^{-\alpha} - \epsilon_i^{-\alpha}}{1 - \epsilon_i^{-\alpha}}$$



(1) Candidate size aware

< For same α >



(2) Expanding range of α

$[0, \infty] \Rightarrow [-1, \infty]$

$$\lim_{\alpha \rightarrow 0} f^*(r, \alpha) = 1 - \frac{\ln r}{\ln \epsilon_i}$$

$[2, 1, 10, \dots, 19, 31, 6]$

f^*

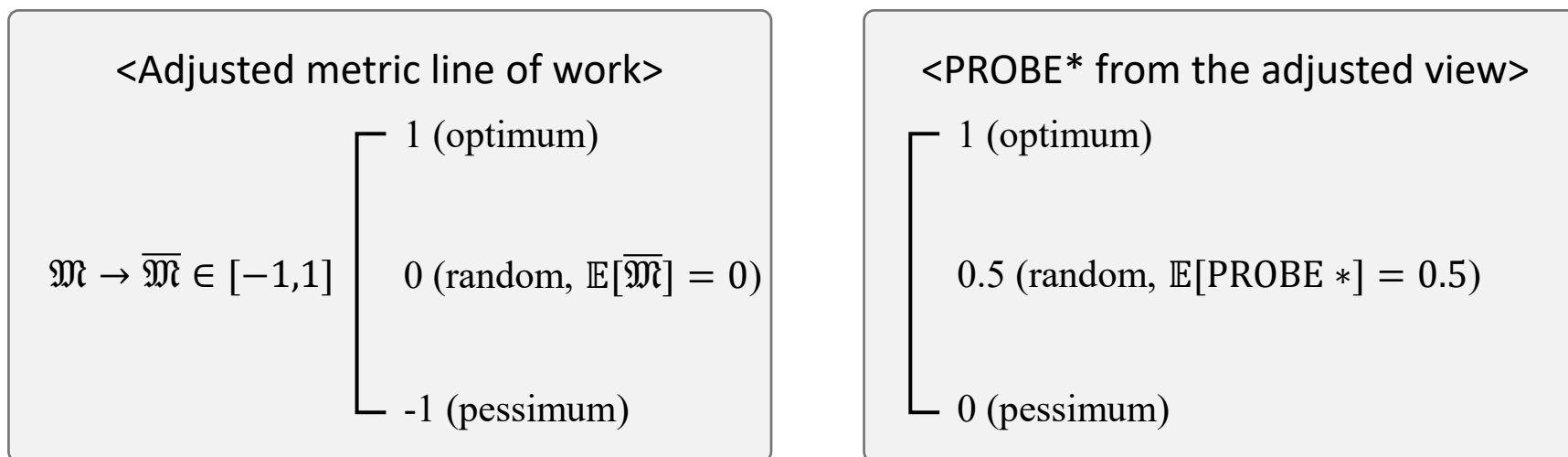
Score vector

Candidate sample of i -th query: ϵ_i

■ (A1) Interpreting Related Works Under PROBE*

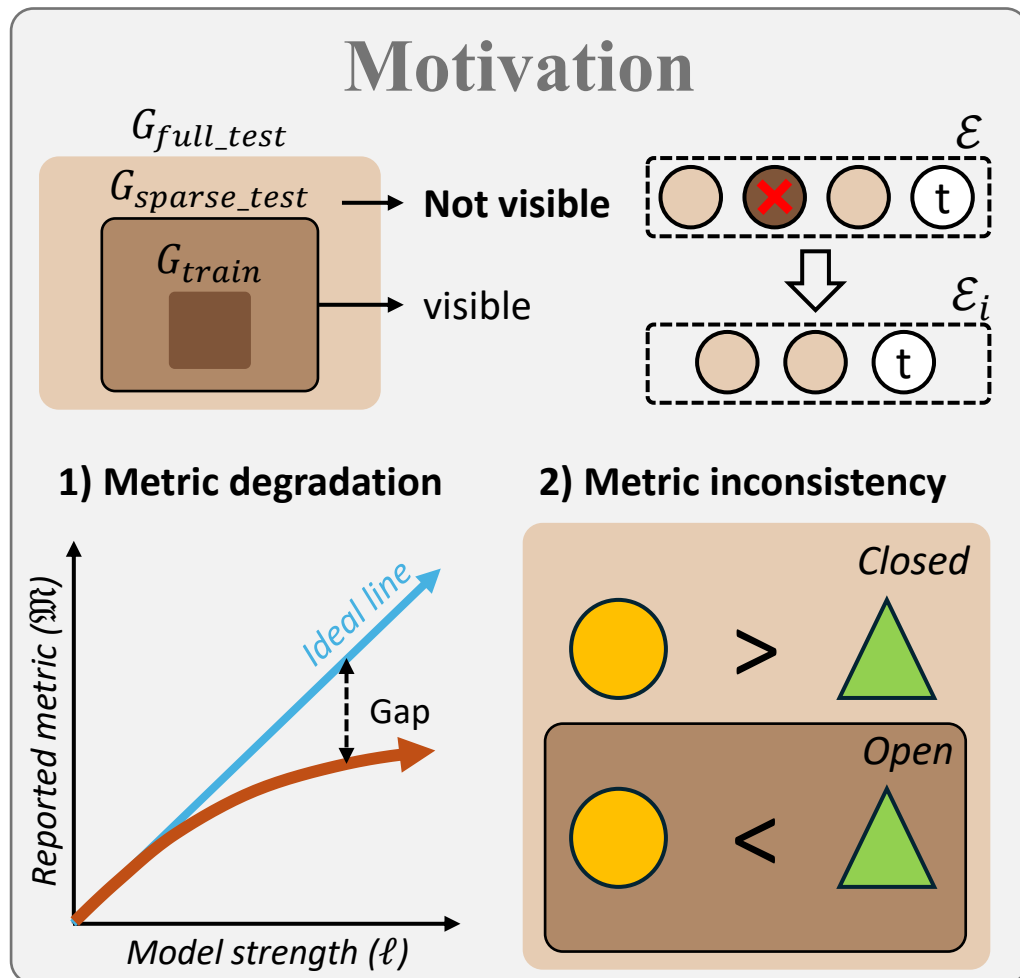
Lemma 1. *Hits@1 is equal to $PROBE^*_{\alpha \rightarrow \infty}$*

Lemma 2. *$PROBE^*_{\alpha=-1}$ is a chance adjusted metric*



PREDICTIVE SHARPNESS

■ (A2) Interpreting Under the Open World (Extending Yang et al.)



Yang's Analysis

$$\frac{d\hat{\mathbb{E}}(\mathfrak{M})}{d\ell} = \frac{1}{l\beta(N+1)} \mathbb{E}_{k \sim \mathcal{B}(N+1, \ell\beta)} g(k)$$
$$g(r) = rf(r)$$

↓ derivative → ↓ degradation & ↓ inconsistency

More credibility to the conclusion

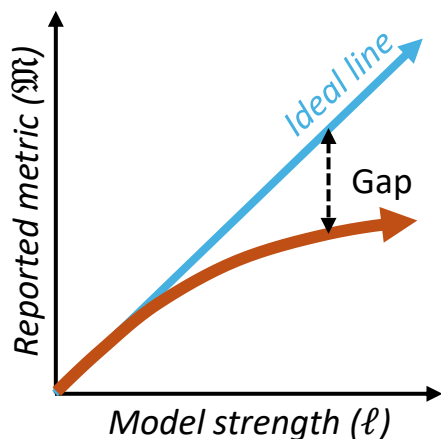
Lemma 3. α strictly decreases PROBE * expectation w. r. t model strength

PREDICTIVE SHARPNESS

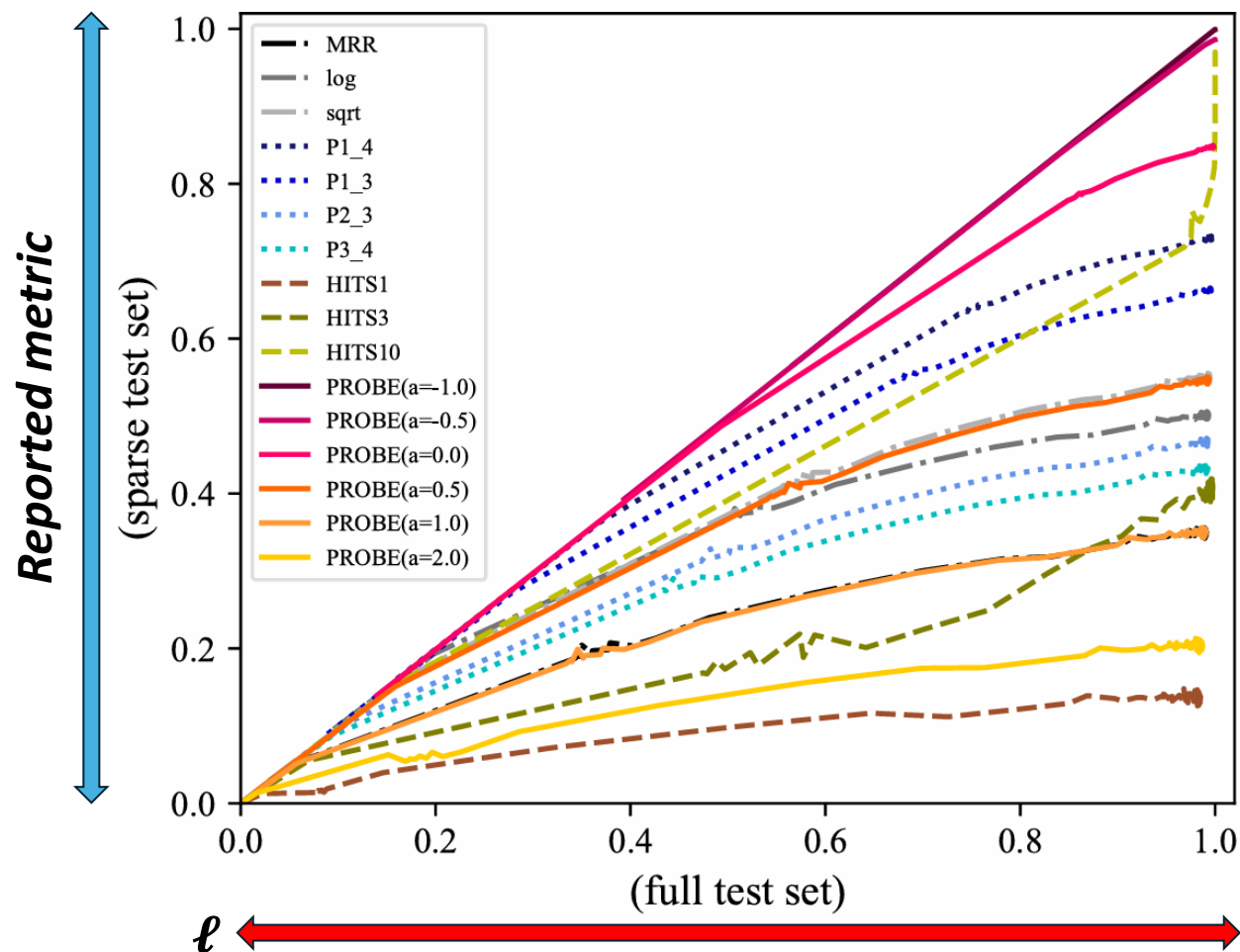
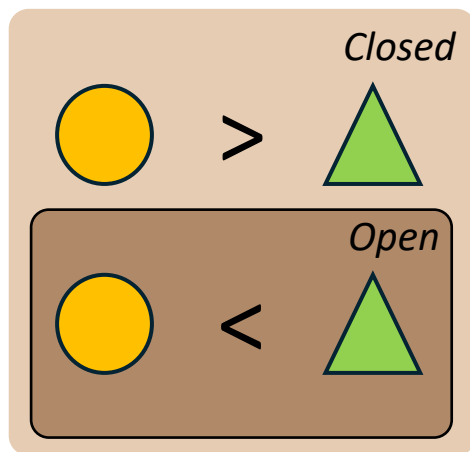
■ (A2) Experiments Under CW and OW

<i>Family-tree Dataset</i>		
# entities	6,004	► Closed-world
# relations	23	► Sparsity controllable (sparsity is set to 75%)
# facts	192,532	

1) Metric degradation



2) Metric inconsistency

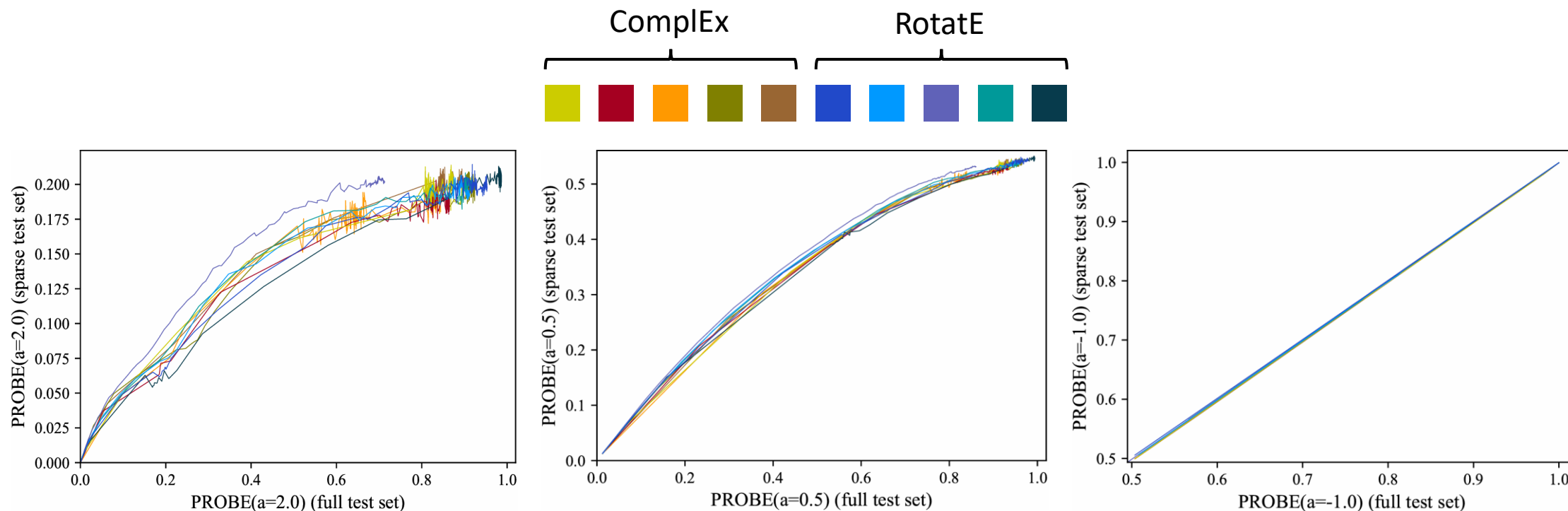


PREDICTIVE SHARPNESS

■ (A2) How Should We View It

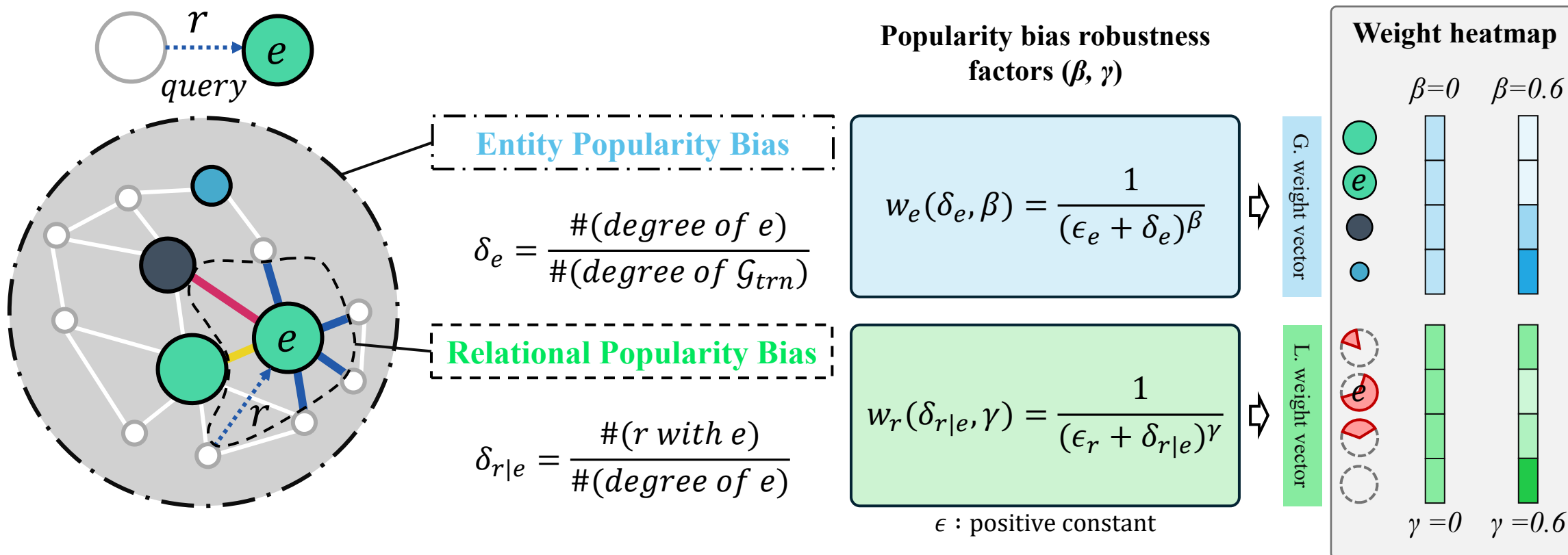
Low sharpness is the best?! Counter example : $\mathfrak{M} = 1$

What we can take How PROBE* reacts to the open world



POPULARITY BIAS ROBUSTNESS

Global Entity Bias & Local Relational Bias



POPULARITY BIAS ROBUSTNESS

Combining Two Weights

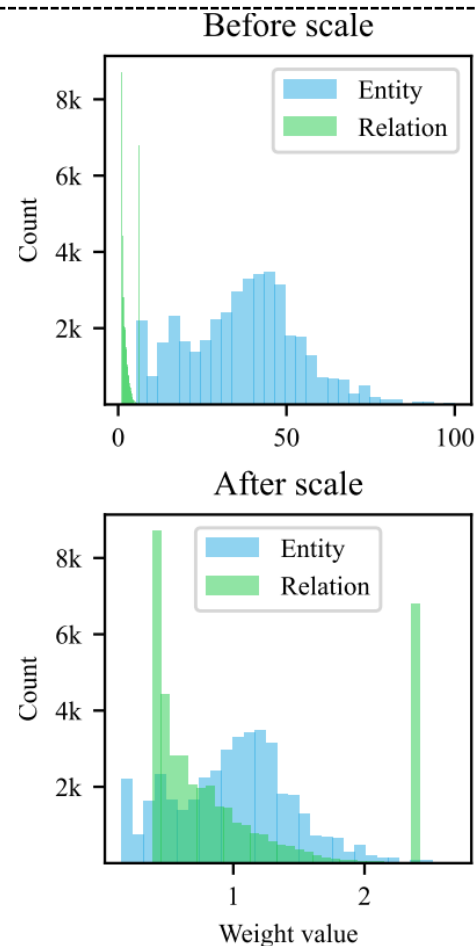
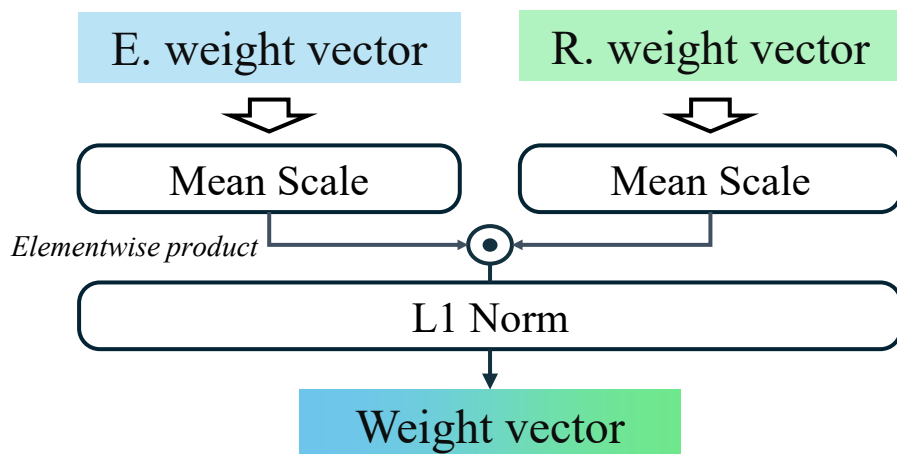
Problem

Two vectors have completely different scale.
One weight can dominate the other.

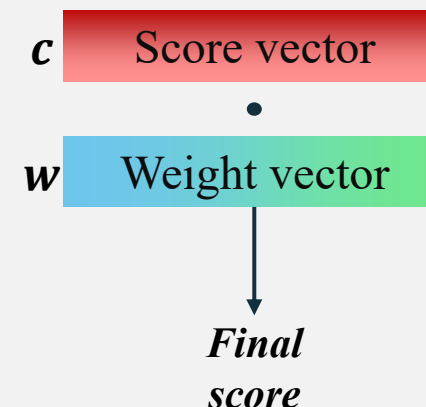


Solution

Mean scale two vectors



Summary of PROBE*



PROBE* is the
dot product between
score and weight vector

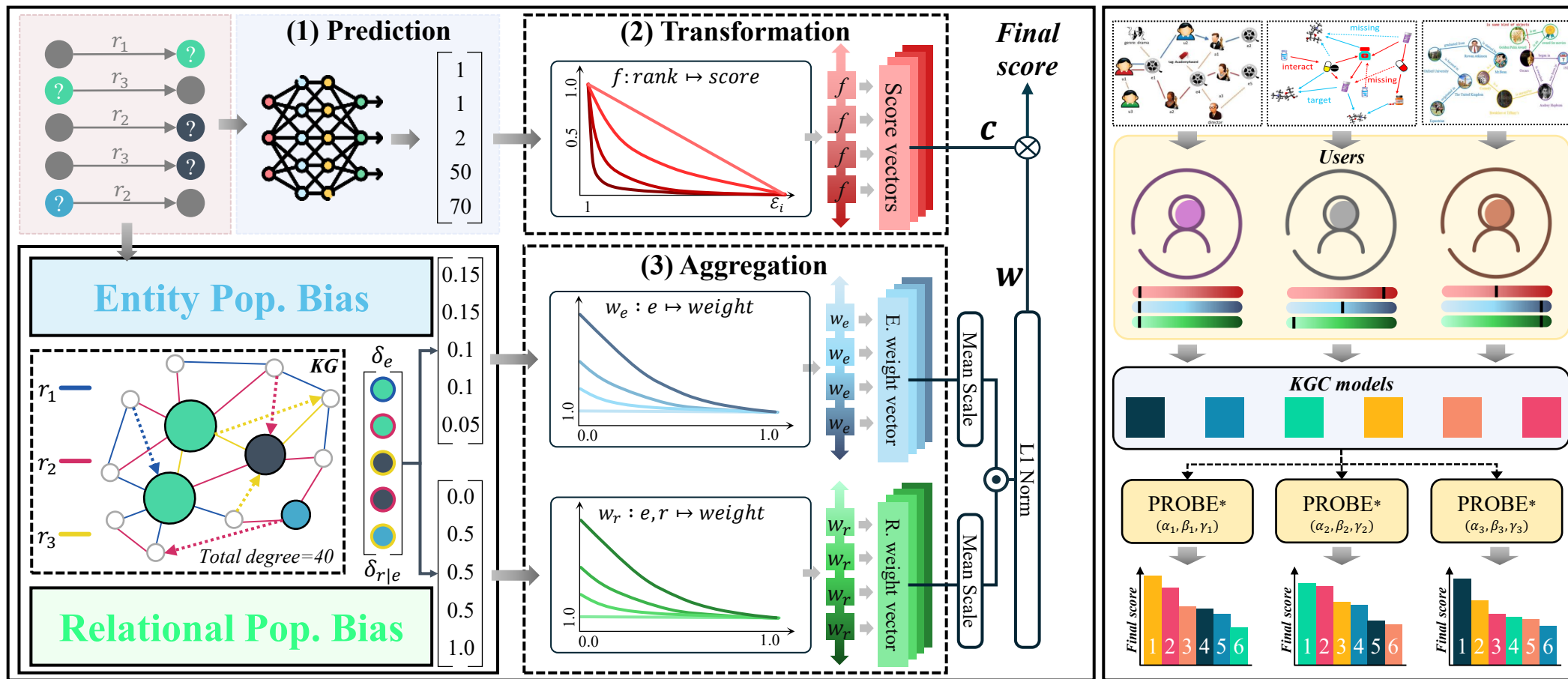
COMBINING PERSPECTIVES

PROBE* Architecture in a Netshell

Background & Motivation

Framework & Analysis

Experiments



COMBINING PERSPECTIVES

■ Desired Properties of Rank Based Metric for KGC

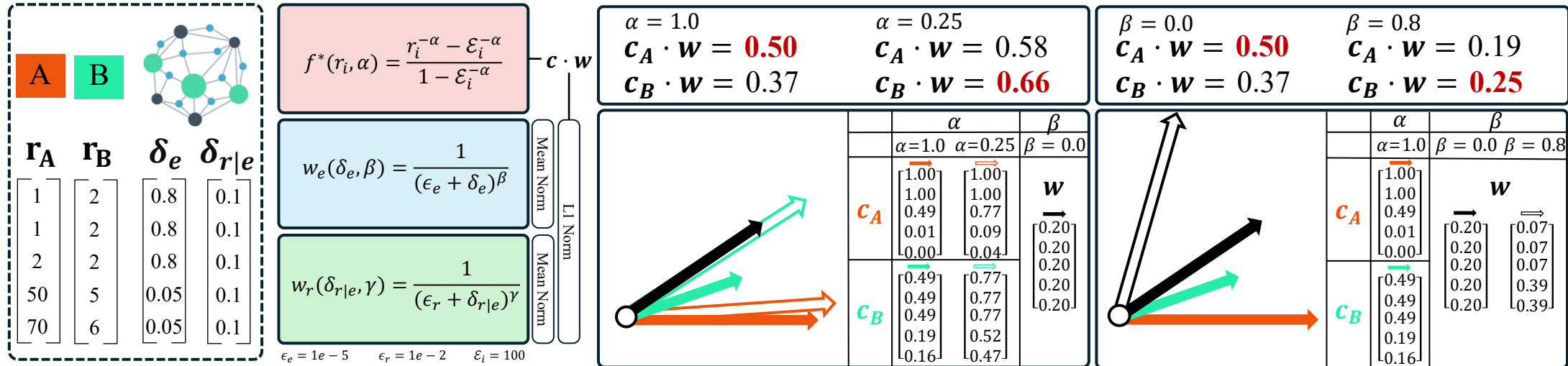
Property	Constraint	MR	MRR	Hits@K	p-MRR	Strat-MRR	WMR	AMRI	PROBE*
Fixed optimum	$\mathfrak{M}(1_{ Q }) = c_{opt}$	✓	✓	✓	✓	✓	✓	✓	✓
Fixed pessimum	$\mathfrak{M}(\mathcal{E}_{ Q }) = c_{pes}$	-	-	✓	-	-	-	✓	✓
Anti-monotonic	$r_i > r_j \rightarrow \mathfrak{M}(r_i) < \mathfrak{M}(r_j)$	-	✓	-	✓	✓	-	✓	✓
Counterpart-aware	$\mathcal{E}_1 \neq \mathcal{E}_2 \rightarrow \mathfrak{M}_1(\mathbf{r}) \neq \mathfrak{M}_2(\mathbf{r})$	-	-	-	-	-	✓	✓	✓
Sharpness-aware	$\alpha \neq \alpha' \rightarrow \mathfrak{M}_\alpha(\mathbf{r}) \neq \mathfrak{M}_{\alpha'}(\mathbf{r})$	-	-	-	✓	-	-	-	✓
Popularity-aware	$w_q \propto \delta(q)$	-	-	-	-	-	-	-	✓

$\mathfrak{M}$	MR	MRR	Hits@K	p-MRR	Strat-MRR	WMR	AMRI	PROBE*
Def	$\frac{1}{ Q } \sum_i r_i$	$\frac{1}{ Q } \sum_i \frac{1}{r_i}$	$\frac{1}{ Q } \sum_i \mathbb{I}[r_i \leq K]$	$\frac{1}{ Q } \sum_i r_i^{-p}$	$\frac{\sum_{rel} w^{rel} strat-MRR_{rel}}{\sum_{rel \in \mathcal{R}} w^{rel}}$	$\exp\left(\frac{\sum_i (\mathcal{E}_i - 1) \ln r_i}{\sum_i (\mathcal{E}_i - 1)}\right)$	$1 - \frac{2 \sum_i (r_i - 1)}{\sum_i (\mathcal{E}_i - 1)}$	$\mathbf{c} \cdot \mathbf{w}$

COMBINING PERSPECTIVES

How Final Scores Change with Perspective Shift

PROBE* is the dot product between score and weight vector



■ Models, Datasets & Implementation Details

● RotatE

■ ComplEx

▲ HousE

▼ TuckER

◆ pLogicNet

◆ RNNLogic

- Low dimension ($d=128$) tuned
- Filtered setting, tie breaking protocol adopted for fair evaluation
- Results are average of 3

$$\alpha = [1.0, 0.5, 0.0, -0.5, -1.0]$$

$$\beta = [0.0, 0.2, 0.4, 0.6, 0.8]$$

$$\gamma = [0.0, 0.2, 0.4, 0.6, 0.8]$$

$$\epsilon_e = 1e - 5$$

$$\epsilon_r = 1e - 2$$

Dataset	$ \mathcal{E} $	$ \mathcal{R} $	$ \mathcal{G}_{trn} $	$ \mathcal{G}_{val} $	$ \mathcal{G}_{tst} $
FB15k237	14,541	237	272,115	17,535	20,466
WN18RR	40,943	11	86,835	3,034	3,134
YAGO3-10	123,182	37	1,079,040	5,000	5,000
Family	3,007	12	23,483	2,038	2,835
UMLS	135	46	1,959	1,306	3,264
Kinship	104	25	3,206	2,137	5,343

■ Perspective Wise Tables

Models	FB15k-237					WN18RR					YAGO3-10				
	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$
RotatE	0.3137	0.4239	0.7124	0.938	0.9844	0.4649	0.5217	0.6945	0.8537	0.9006	<u>0.4729</u>	<u>0.5625</u>	<u>0.7966</u>	0.9554	0.9812
ComplEx	0.3135	0.4274	<u>0.7209</u>	<u>0.9424</u>	<u>0.9854</u>	0.4752	0.5183	0.661	0.815	0.873	0.4447	0.5426	<u>0.7957</u>	0.9577	0.9807
HousE	0.3073	0.41	0.691	0.9244	0.9778	0.4184	0.4913	<u>0.6991</u>	<u>0.8823</u>	0.9346	0.4538	0.5297	0.75	0.9267	0.9647
TuckER	0.353	0.4608	0.7379	0.9479	0.9884	0.4625	0.4999	0.6368	0.7961	0.8583	0.5654	0.6319	0.8176	0.9561	<u>0.9814</u>
pLogicNet	<u>0.3241</u>	<u>0.4311</u>	0.7149	0.9386	0.9847	0.4957	0.5672	0.7593	0.9002	<u>0.9292</u>	0.4535	0.5467	0.7927	<u>0.9576</u>	0.9834
RNNLogic	0.3197	0.4218	0.6969	0.9213	0.9723	<u>0.4807</u>	<u>0.5225</u>	0.6567	0.7929	0.8401	-	-	-	-	-

Models	Family					UMLS					Kinship				
	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$
RotatE	0.9478	0.9639	0.9832	0.9921	0.9938	0.7782	0.8369	0.9056	0.9556	0.9776	0.6318	0.7224	0.8365	0.9273	0.9717
ComplEx	0.9431	0.9616	0.9835	0.9941	0.9966	0.5993	0.6796	0.79	0.885	0.9362	0.5416	0.648	0.787	0.9015	0.9596
HousE	<u>0.9757</u>	<u>0.9818</u>	<u>0.9893</u>	0.9938	0.9953	<u>0.8354</u>	<u>0.8775</u>	<u>0.9271</u>	<u>0.9637</u>	<u>0.9802</u>	<u>0.6808</u>	<u>0.762</u>	<u>0.8621</u>	<u>0.94</u>	0.977
TuckER	0.9784	0.9832	0.9895	<u>0.9939</u>	<u>0.9959</u>	0.8439	0.8849	0.9323	0.9669	0.9825	0.7067	0.7808	0.8716	0.9421	<u>0.9759</u>
pLogicNet	0.8119	0.86	0.9355	0.9837	0.9939	0.5781	0.6608	0.7748	0.8764	0.9346	0.3771	0.4501	0.5628	0.6803	0.7641
RNNLogic	0.9737	0.9781	0.9839	0.9882	0.9901	0.7942	0.8455	0.9068	0.9526	0.9739	0.6417	0.7289	0.8394	0.9277	0.9712

EXPERIMENTS

Case Study : Predictive Sharpness

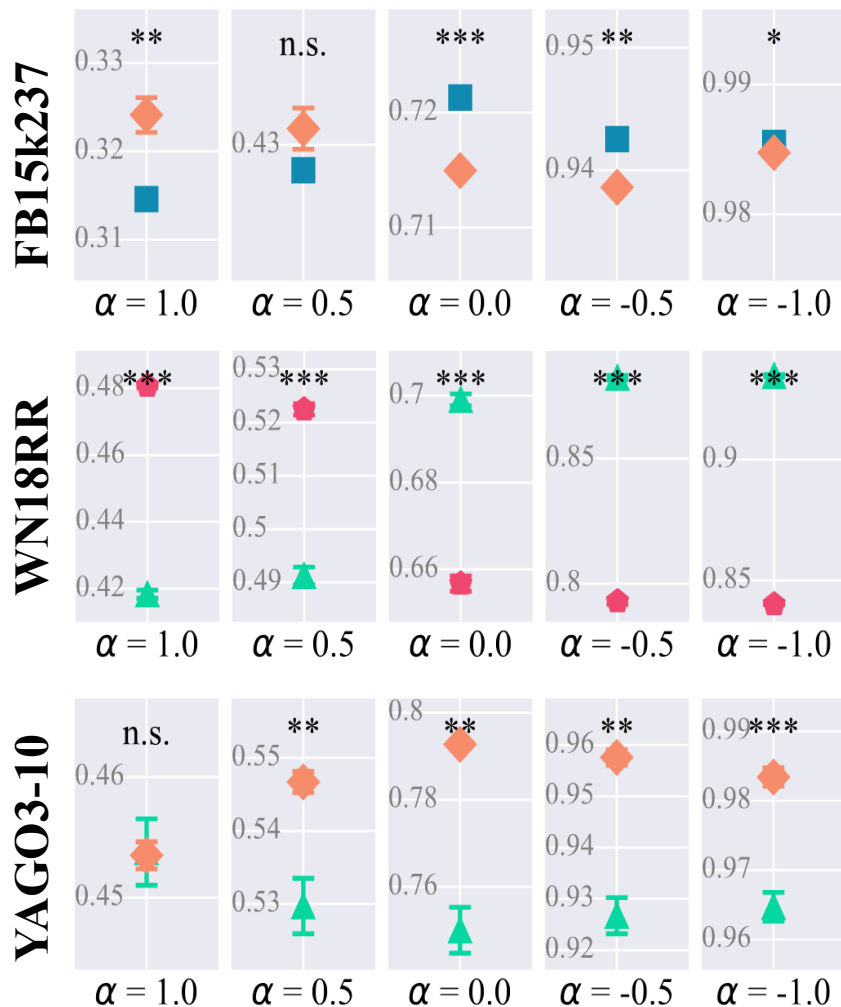
Results with
statistical
significance

$p \geq 5e-2$ (n.s.)

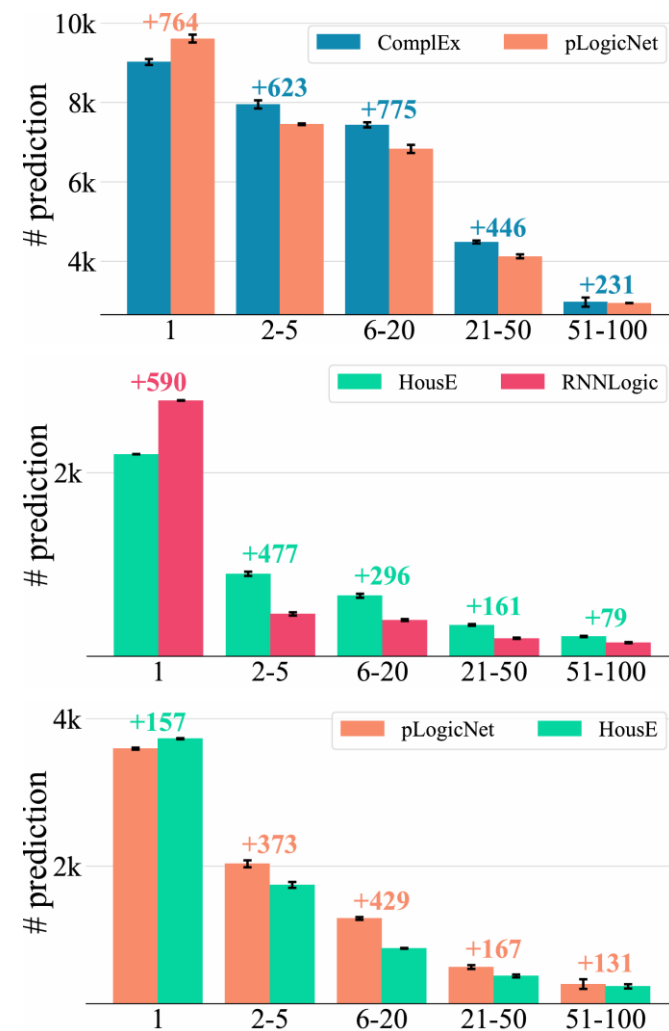
$p < 5e-2$ (*)

$p < 1e-2$ (**)

$p < 1e-3$ (***)



Why?
⇒



EXPERIMENTS

■ Perspective Wise Tables

Models	FB15k-237					WN18RR					YAGO3-10				
	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$
RotatE	0.314	0.2633	0.2286	0.2045	0.1873	0.4641	0.4485	0.4318	0.4141	0.3957	0.4743	0.4288	0.3976	0.3747	0.3572
ComplEx	0.3146	0.2662	0.2319	0.2072	0.1886	0.4754	0.4574	0.4381	0.4178	0.3971	0.4459	0.418	0.39	0.3665	0.3475
HousE	0.3073	0.2533	0.2167	0.1916	0.1737	0.4184	0.4038	0.3897	0.3754	0.3607	0.4538	0.4104	0.3793	0.3556	0.337
Tucker	0.353	0.2948	0.2544	0.226	0.2053	0.4625	0.4426	0.4225	0.4022	0.3816	0.5654	0.5178	0.4842	0.4589	0.4391
pLogicNet	0.3241	0.2772	0.2458	0.2247	0.2103	0.4957	0.4657	0.4388	0.4143	0.3916	0.4535	0.4091	0.3791	0.3575	0.3414
RNNLogic	0.3197	0.2641	0.2266	0.2011	0.1831	0.4807	0.4572	0.4345	0.4121	0.3897	-	-	-	-	-

Models	Family					UMLS					Kinship				
	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$
RotatE	0.9487	0.9404	0.9278	0.9084	0.879	0.7784	0.7625	0.7363	0.6845	0.5736	0.6336	0.633	0.6325	0.6319	0.6313
ComplEx	0.9434	0.939	0.9307	0.9158	0.8908	0.6046	0.6047	0.601	0.5875	0.5522	0.5394	0.5389	0.5383	0.5377	0.5372
HousE	0.9757	0.9689	0.9572	0.938	0.9078	0.8354	0.8229	0.7996	0.7482	0.6315	0.6808	0.6802	0.6795	0.6789	0.6783
Tucker	0.9784	0.9719	0.9608	0.9426	0.9138	0.8439	0.8318	0.8129	0.779	0.7143	0.7067	0.7058	0.7049	0.7041	0.7032
pLogicNet	0.8119	0.8026	0.7896	0.7714	0.7462	0.5781	0.5786	0.5741	0.5553	0.5009	0.3771	0.3768	0.3765	0.3762	0.3759
RNNLogic	0.9737	0.9647	0.9506	0.9286	0.8957	0.7942	0.7793	0.7541	0.7037	0.5967	0.6417	0.6408	0.6398	0.6388	0.6379

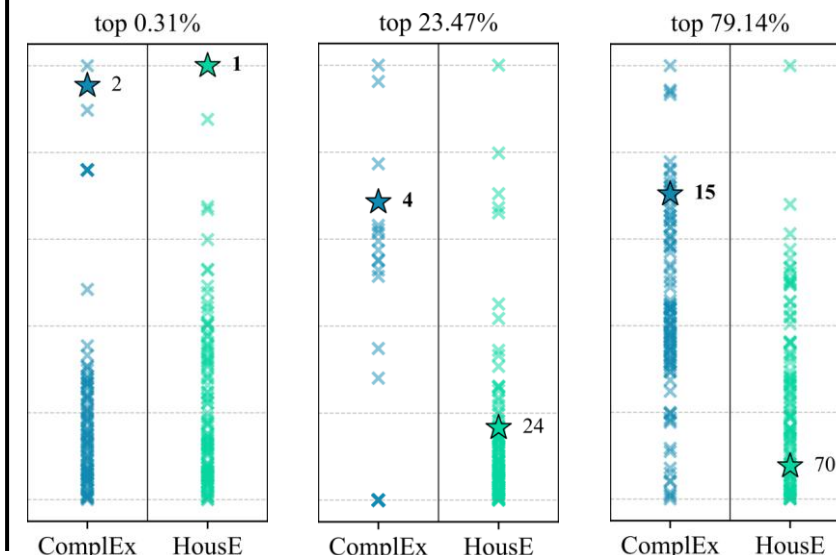
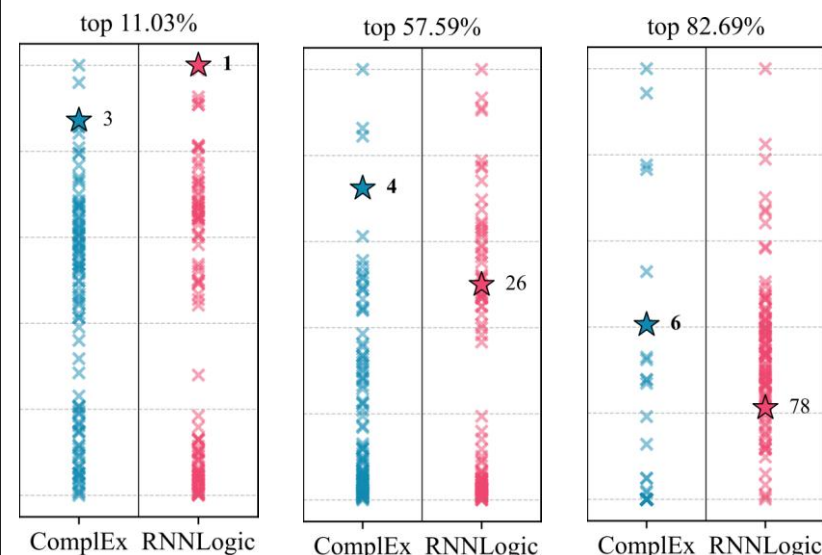
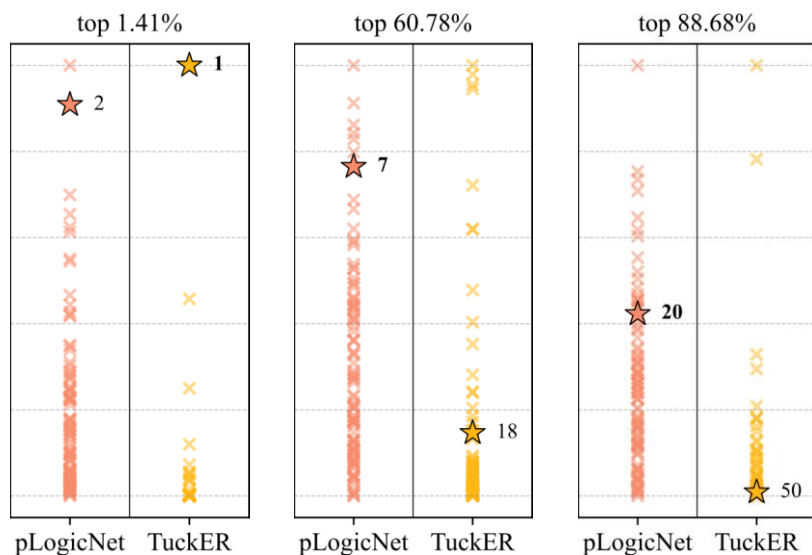
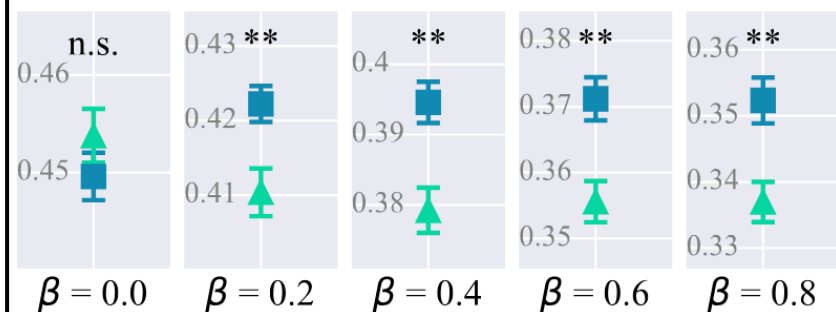
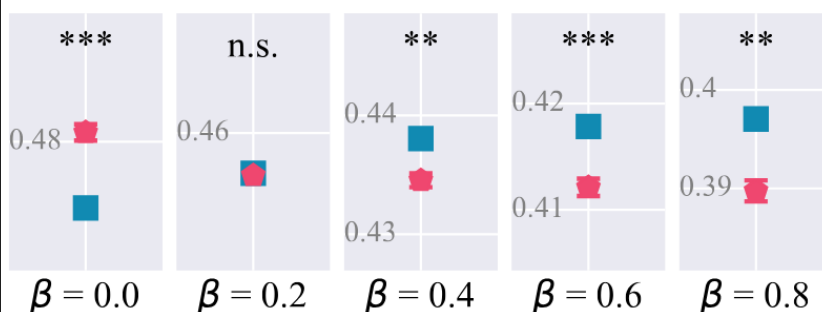
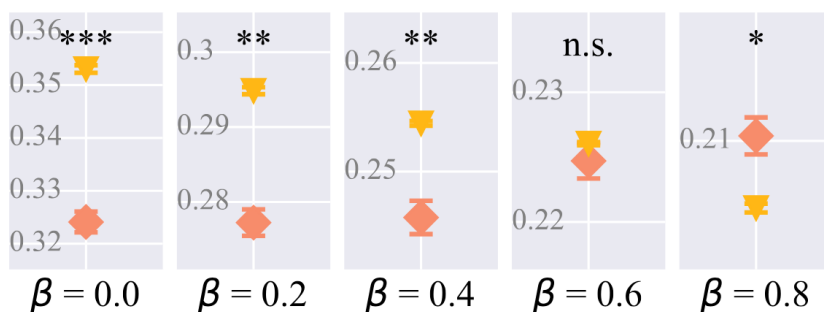
EXPERIMENTS

Case Study (Entity Bias) : Top-100 Prediction Scores

FB15k237

WN18RR

YAGO3-10



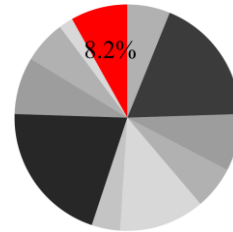
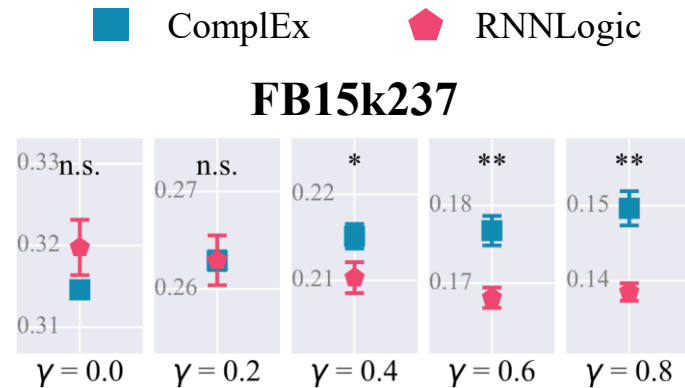
EXPERIMENTS

■ Perspective Wise Tables

Models	FB15k-237					WN18RR					YAGO3-10				
	$\gamma=0.0$	$\gamma=0.2$	$\gamma=0.4$	$\gamma=0.6$	$\gamma=0.8$	$\gamma=0.0$	$\gamma=0.2$	$\gamma=0.4$	$\gamma=0.6$	$\gamma=0.8$	$\gamma=0.0$	$\gamma=0.2$	$\gamma=0.4$	$\gamma=0.6$	$\gamma=0.8$
RotatE	0.314	0.2603	0.2104	0.1702	0.1417	0.4641	0.3854	<u>0.2972</u>	<u>0.2244</u>	<u>0.1777</u>	<u>0.4743</u>	<u>0.4347</u>	<u>0.3698</u>	<u>0.2856</u>	<u>0.2031</u>
ComplEx	0.3146	<u>0.2629</u>	0.2151	0.1768	0.1496	0.4754	<u>0.3913</u>	0.2969	0.2188	0.1685	0.4496	0.4156	0.3569	0.279	0.2016
HousE	0.3073	0.2535	0.2032	0.1623	0.1331	0.4184	0.3483	0.2704	0.2065	0.1657	0.4538	0.4115	0.3412	0.2497	0.1599
TuckER	0.353	0.2961	0.2433	0.201	0.1713	0.4625	0.3735	0.2738	0.1915	0.1386	0.5654	0.5187	0.4414	0.3409	0.2422
pLogicNet	<u>0.3241</u>	<u>0.2726</u>	<u>0.224</u>	<u>0.1843</u>	<u>0.1558</u>	0.4957	0.4118	0.3198	0.245	0.1975	0.4535	0.417	0.3565	0.2776	0.1997
RNNLogic	<u>0.3197</u>	0.2629	0.2103	0.1681	0.1385	<u>0.4807</u>	0.3897	0.2875	0.2031	0.1487	-	-	-	-	-

Models	Family					UMLS					Kinship				
	$\gamma=0.0$	$\gamma=0.2$	$\gamma=0.4$	$\gamma=0.6$	$\gamma=0.8$	$\gamma=0.0$	$\gamma=0.2$	$\gamma=0.4$	$\gamma=0.6$	$\gamma=0.8$	$\gamma=0.0$	$\gamma=0.2$	$\gamma=0.4$	$\gamma=0.6$	$\gamma=0.8$
RotatE	0.9487	0.9484	0.9474	0.9458	0.9439	0.7784	0.771	<u>0.7585</u>	0.7418	0.7227	0.6336	0.6271	0.6193	0.6101	0.5996
ComplEx	0.9434	0.9462	0.9485	0.9503	0.9514	0.6046	0.6131	0.6179	0.6193	0.6178	0.5394	0.5354	0.5304	0.5242	0.5169
HousE	<u>0.9757</u>	<u>0.9738</u>	<u>0.971</u>	<u>0.9678</u>	<u>0.9647</u>	<u>0.8354</u>	0.8269	0.8132	0.7952	0.7747	<u>0.6808</u>	<u>0.6705</u>	<u>0.6584</u>	<u>0.6444</u>	0.6287
TuckER	0.9784	0.9761	0.9731	0.9696	0.9663	0.8439	<u>0.8262</u>	<u>0.803</u>	<u>0.7755</u>	<u>0.7458</u>	0.7067	0.6905	0.6714	0.6495	<u>0.6251</u>
pLogicNet	0.8119	0.7989	0.7845	0.7704	0.7583	0.5781	0.5615	0.5405	0.5165	0.4917	0.3771	0.3798	0.3822	0.3841	0.3854
RNNLogic	0.9737	0.9699	0.9649	0.9592	0.9537	0.7942	0.7776	0.7554	0.7288	0.7001	0.6417	0.6222	0.5993	0.5732	0.5444

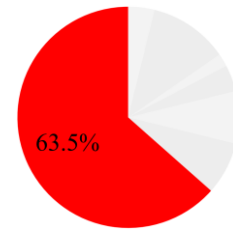
Case Study (Relational Bias) : Relation Pie Chart



The Last King of Scotland, ./genre, docudrama

The Last King of Scotland, ./genre, thriller

The Last King of Scotland, ./genre, war film



Magic Mike, ./film_release_region, United Kingdom

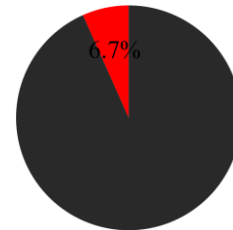
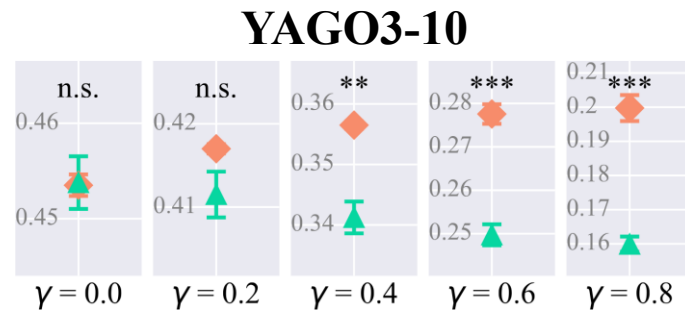
Magic Mike, ./film_release_region, Finland

Rank

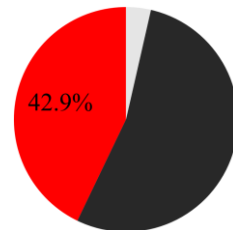


<u>39</u>	188
<u>572</u>	798
<u>80</u>	751

◆ pLogicNet ▲ HousE



Bridgeport_Sound_Tigers,
isLocatedIn, Fairfield_County,_Connecticut



Massimo_Maccarone,
playsFor, Italy_national_under-20_football_team

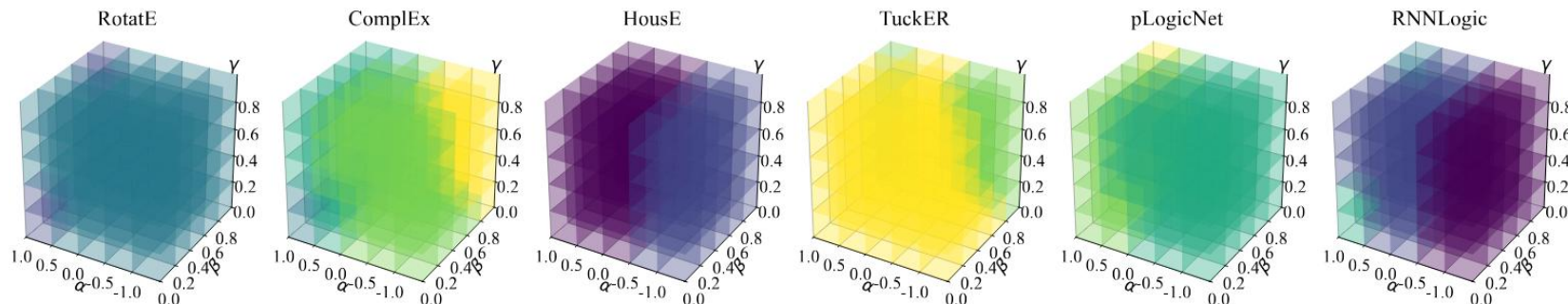
◆	▲
<u>164</u>	2610

50	<u>5</u>
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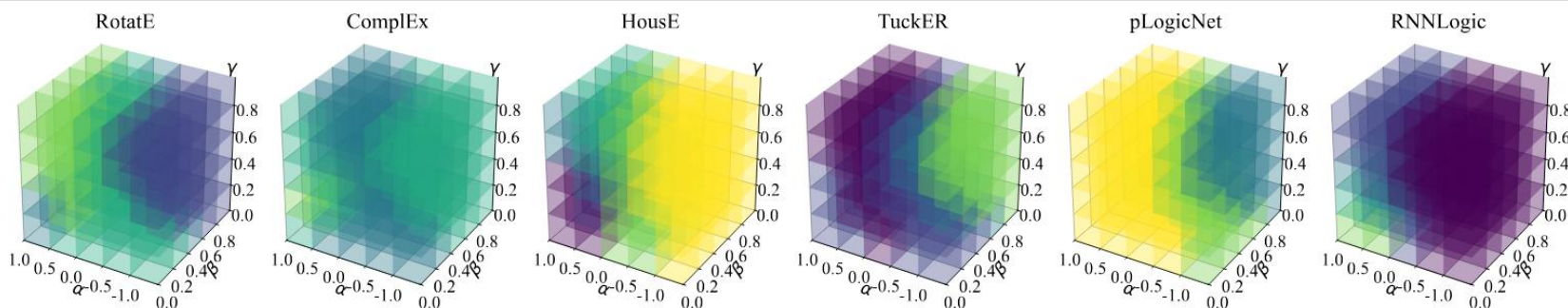
EXPERIMENTS

■ Tri-Perspective Cube Plot

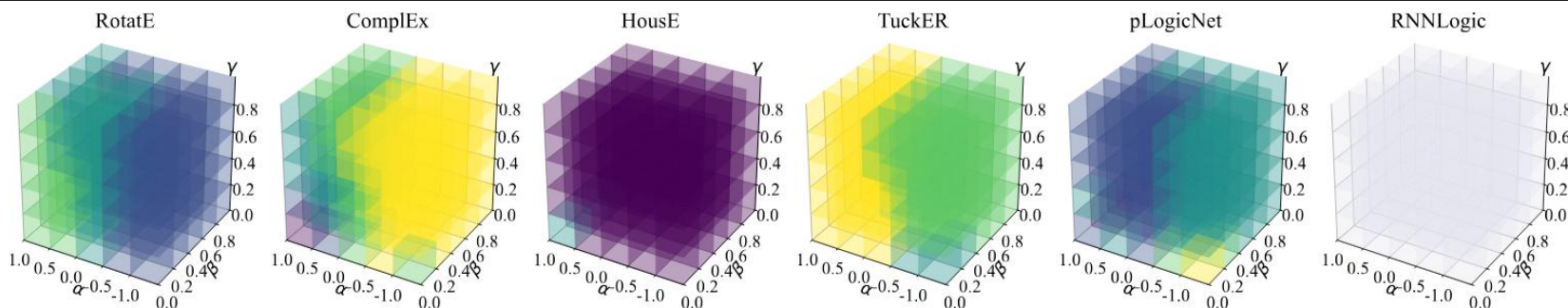
FB15k237



WN18RR



YAGO3-10



Conclusion & Future Works

Background & Motivation

Framework & Analysis

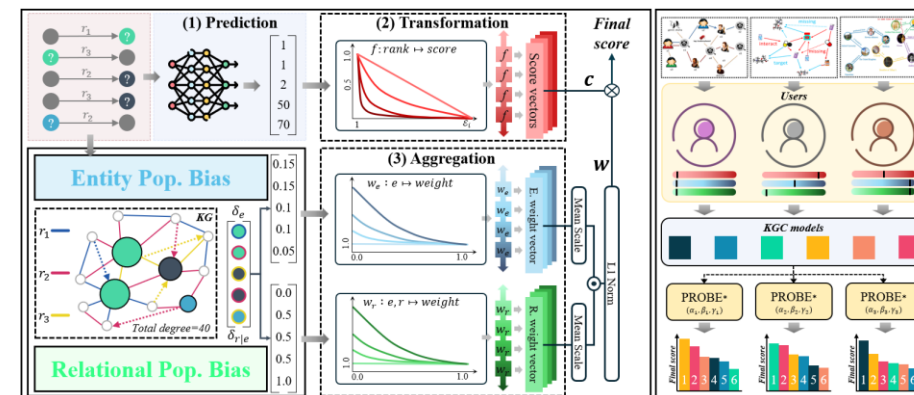
Experiments

Motivation for Perspective-wise Metric

- Different objectives & use-cases of KGC models

Generalized Framework (PROBE*)

- Predictive sharpness : degree of penalization
- Popularity bias robustness : degree of robustness on low popular entities / pairs

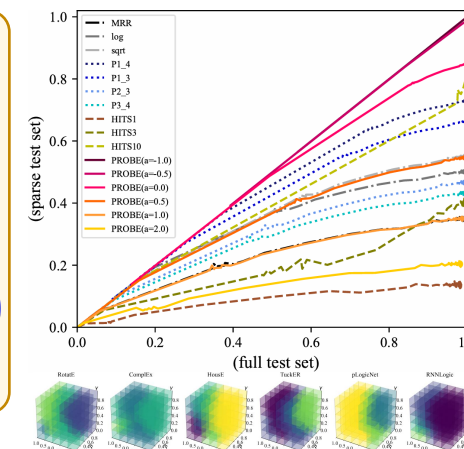
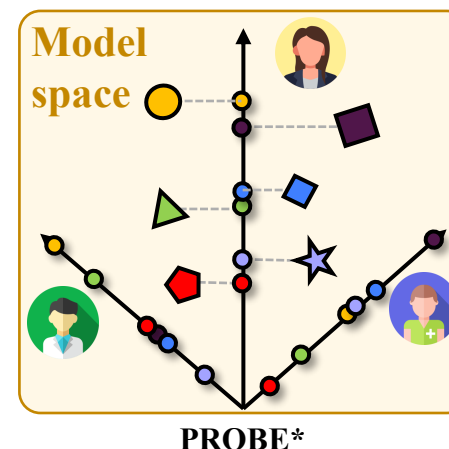


Comprehensive Analysis

- Theoretically analyze relationships between related works

Experiments & Case Studies

- Validated evaluation results of PROBE*



Conclusion & Future Works

■ Impact of Popularity Bias on the OW

- ☐ Synthesized family KG has near uniform degree distribution
- ☐ Investigating RA under more realistic KG (power law)

■ Deeper Analysis on RA

- ☐ A perspective constraint imposed to the KG itself
- ☐ Need theoretical justifications other than heuristic weight functions

■ Are There Method-Wise Strength or Weakness?

- ☐ Do some methods have advantages over these perspectives? If so, Why?

Q&A

■ A. Proof of Lemma 1 and 2

Lemma 1. *Hits@1 is equal to $PROBE_{\alpha \rightarrow \infty}$*

$$PROBE_{\alpha \rightarrow \infty} = \lim_{\alpha \rightarrow \infty} \frac{r^{-\alpha} - \mathcal{E}_i^{-\alpha}}{1 - \mathcal{E}_i^{-\alpha}}$$
$$= \begin{cases} 1 & \text{if } r = 1 \\ 0 & \text{if } r > 1 \end{cases}$$

Lemma 2. *$PROBE_{\alpha=-1}$ is a chance adjusted metric*

Let's assume $r_i \sim \text{Uniform}(\{1, 2, \dots, \mathcal{E}_i\})$ where $\mathbb{E}[r_i] = \frac{\mathcal{E}_i+1}{2}$

$$\begin{aligned} \mathbb{E}[PROBE_{\alpha=-1}] &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \frac{r_i - \mathcal{E}_i}{1 - \mathcal{E}_i}\right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}\left[\frac{r_i - \mathcal{E}_i}{1 - \mathcal{E}_i}\right] \\ &= \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{E}[r_i] - \mathcal{E}_i}{1 - \mathcal{E}_i} \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1 - \mathcal{E}_i}{2(1 - \mathcal{E}_i)} \\ &= \frac{1}{2}. \end{aligned}$$

■ B. Proof of Lemma 3 (1)

Lemma 3. α strictly decreases $PROBE$ * expectation w.r.t model strength

We first discuss the bounds of variable k and $\hat{N}$ (absolute lower bound of rank). According to Corollary 4.1 from Yang et al. [32], the derivative of metric expectation w.r.t model strength is

$$\frac{d \hat{\mathbb{E}}(\mathfrak{M})}{d\ell} = \frac{1}{\ell \beta(N+1)} \mathbb{E}_{k \sim \mathcal{B}(N+1, \ell \beta)} g(k). \quad (5)$$

The random variant $\mathfrak{m}$ is the number of missing answers from the number of full answers N which follows the binomial distribution $\mathfrak{m} \sim \mathcal{B}(N, \beta)$. Thus, the number of filtered entity is $N - \mathfrak{m}$ which results in $\hat{N} = N_e - (N - \mathfrak{m})$. Here, $g(0) = 0$ by Yang et al. [32], which leads to the domain bound $1 \leq k \leq N+1$ rather than $0 \leq k \leq N+1$. By following the statement $N/N_e < 10\%$ in almost all queries [32], we can expand the domain bound to $1 \leq k \leq N+1 < \hat{N} \leq N_e$.

The transform function and function $g_\alpha(k)$ is denoted as follows.

$$f_\alpha^*(k) := \frac{k^{-\alpha} - \hat{N}^{-\alpha}}{1 - \hat{N}^{-\alpha}} \quad g_\alpha(k) := k f_\alpha^*(k), \quad 1 \leq k \leq N+1.$$

Note that we don't consider $\alpha = 0$ since f_0^* is not strictly defined.

■ B. Proof of Lemma 3 (2)

Since both the denominator and the binomial distribution from 5 doesn't rely on α , the sign of the partial derivative solely relies on $g_\alpha(k)$. Thus we can reduce the question to *prove $g_\alpha(k)$ strictly decreases respect to α .*

$$\frac{\partial}{\partial \alpha} g_\alpha(k) = \frac{\partial}{\partial \alpha} k f_\alpha^*(k) \quad (6)$$

$$\propto \frac{\partial}{\partial \alpha} f_\alpha^*(k) \quad (7)$$

$$= \hat{N}^\alpha k^{-\alpha} (\hat{N}^\alpha - 1) (\ln \hat{N} - \ln k) - \hat{N}^\alpha (\hat{N}^\alpha k^{-\alpha} - 1) \ln \hat{N} \quad (8)$$

$$= \hat{N}^\alpha k^{-\alpha} ((k^\alpha - 1) \ln \hat{N} + (1 - \hat{N}^\alpha) \ln k) \quad (9)$$

$$= \hat{N}^\alpha k^{-\alpha} F_\alpha(k). \quad (10)$$

Since $\hat{N}^\alpha k^{-\alpha} > 0$, we now have to prove that $F_\alpha(k) < 0$

$$\frac{\partial}{\partial \alpha} F_\alpha(k) = (k^\alpha - \hat{N}^\alpha) \ln k \ln \hat{N}.$$

Due to $k < \hat{N}$,

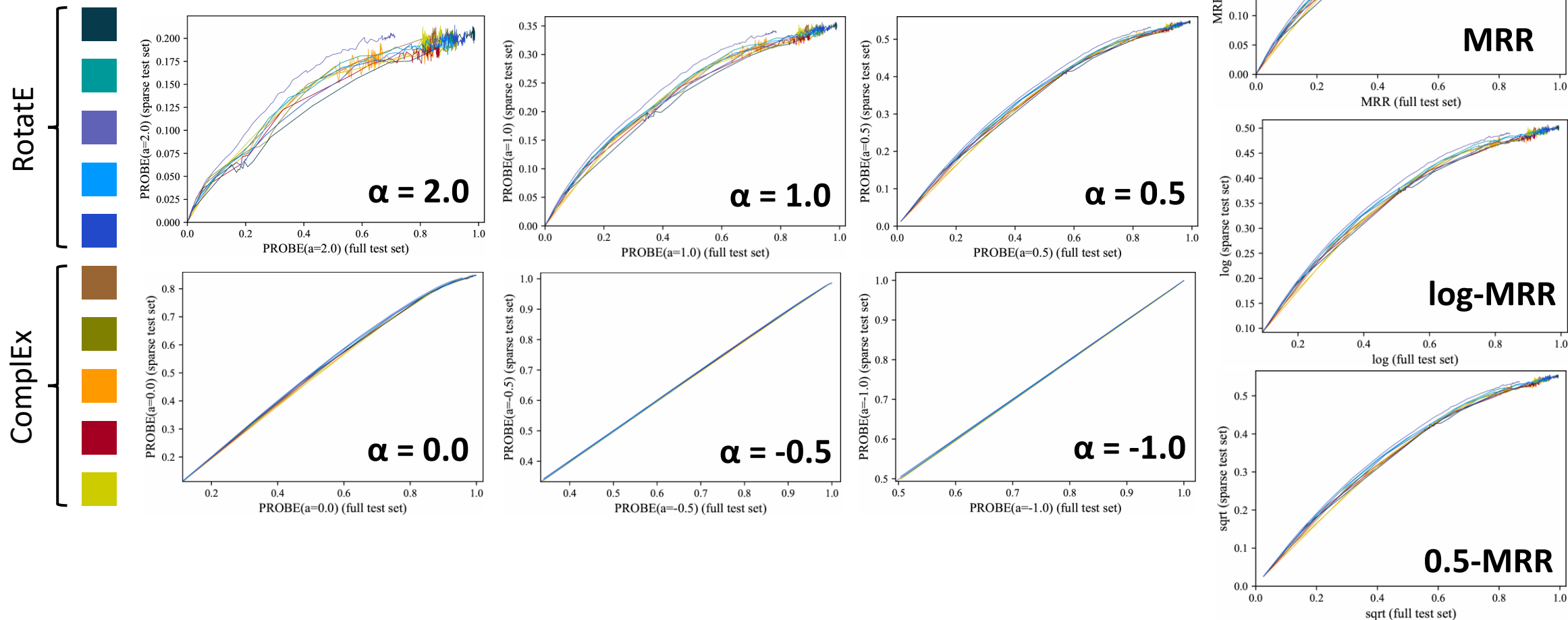
Case 1 $\alpha > 0$. $\frac{\partial}{\partial \alpha} F_\alpha(k) < 0$,

Case 2 $\alpha < 0$. $\frac{\partial}{\partial \alpha} F_\alpha(k) > 0$,

and since $\lim_{\alpha \rightarrow 0} F_\alpha(k) = 0$, we can conclude $F_\alpha(k) < 0$ for $\alpha \in \mathbb{R} \setminus \{0\}$ and thus, $\frac{\partial}{\partial \alpha} \left(\frac{d\hat{\mathbb{E}}(PROBE_\alpha)}{d\ell} \right) < 0$.

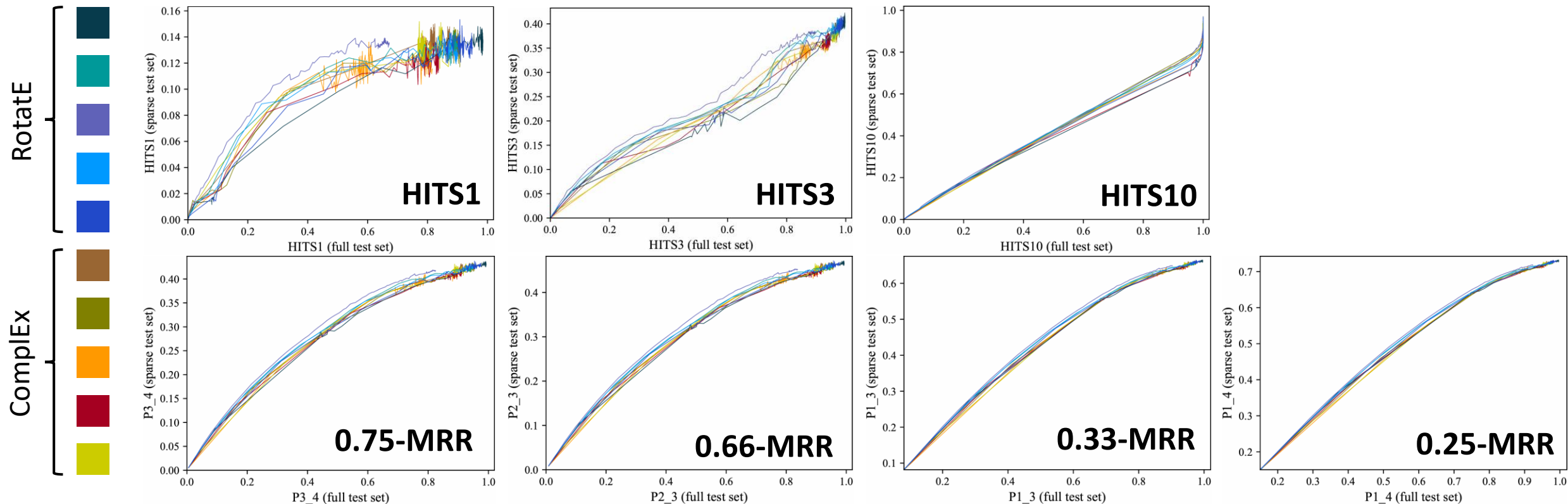
Appendix

■ C. More Plots Under the Open World

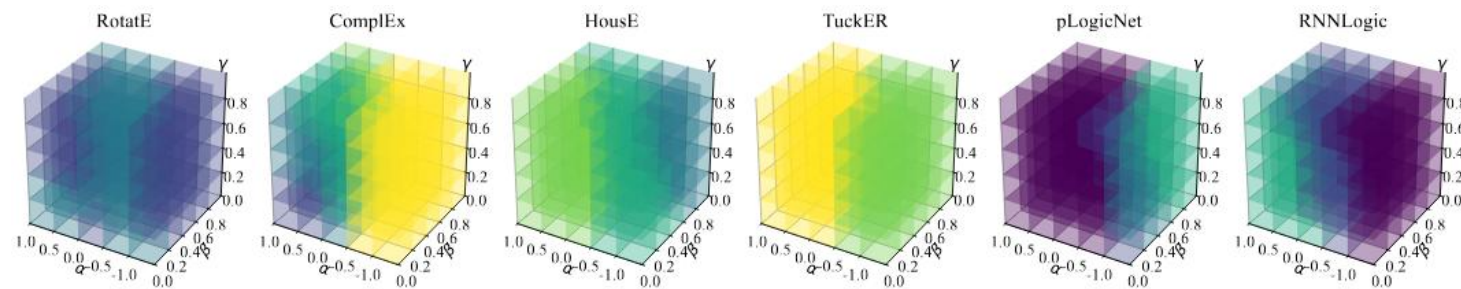


Appendix

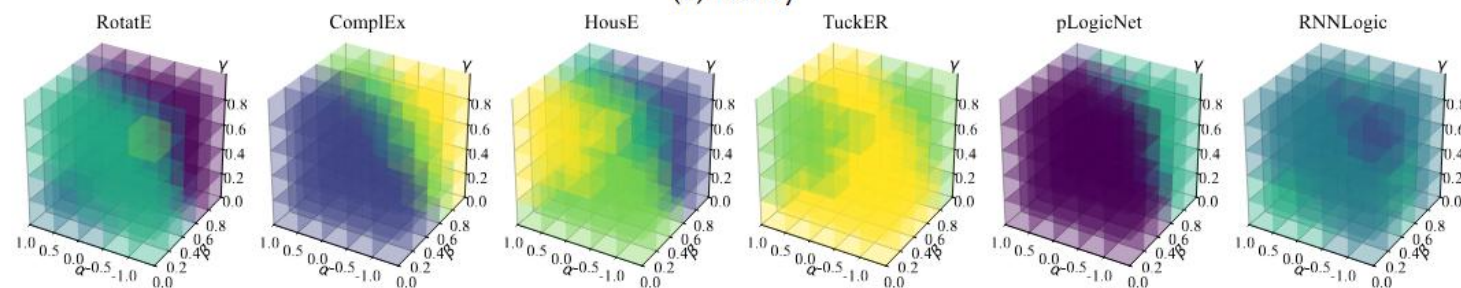
■ C. More Plots Under the Open World



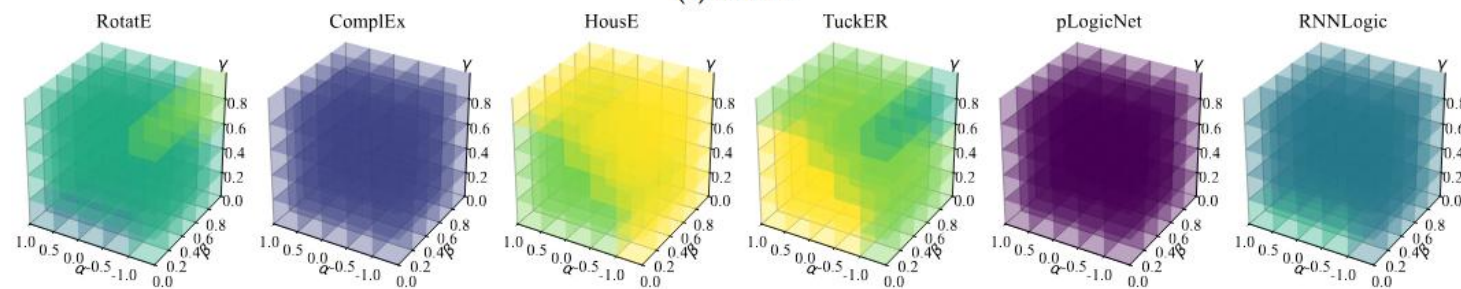
■ D. More Tri-Perspective Plots



(d) Family



(e) UMLS



(f) Kinship